

Stable Diffusion

MIRPR – 2024

Alexandru Manole



Introduction - Rabbit Hole



“THE BAD ARTISTS
IMITATE, THE GREAT
ARTISTS STEAL.”

~~PABLO PICASSO~~

~~BANKSY~~

A MANOLE

Rabbit Hole Progress Tracking

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Wild Mentimeter
appeared!

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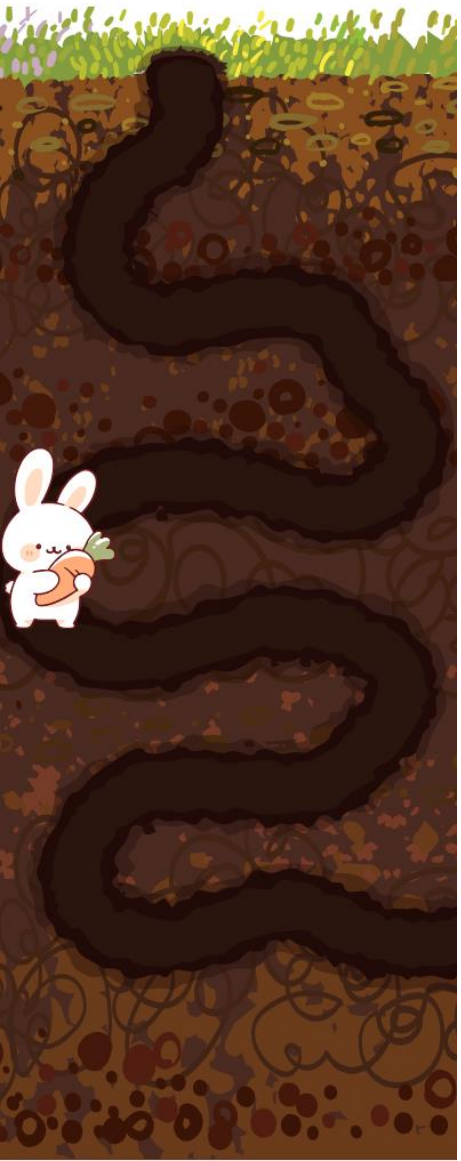
Enter the code

6718 8786



Or use QR code

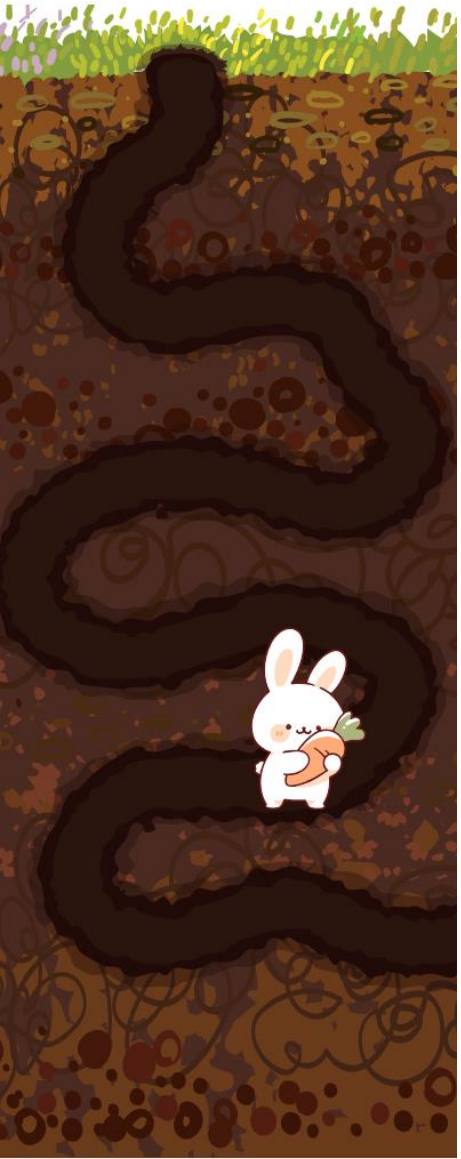




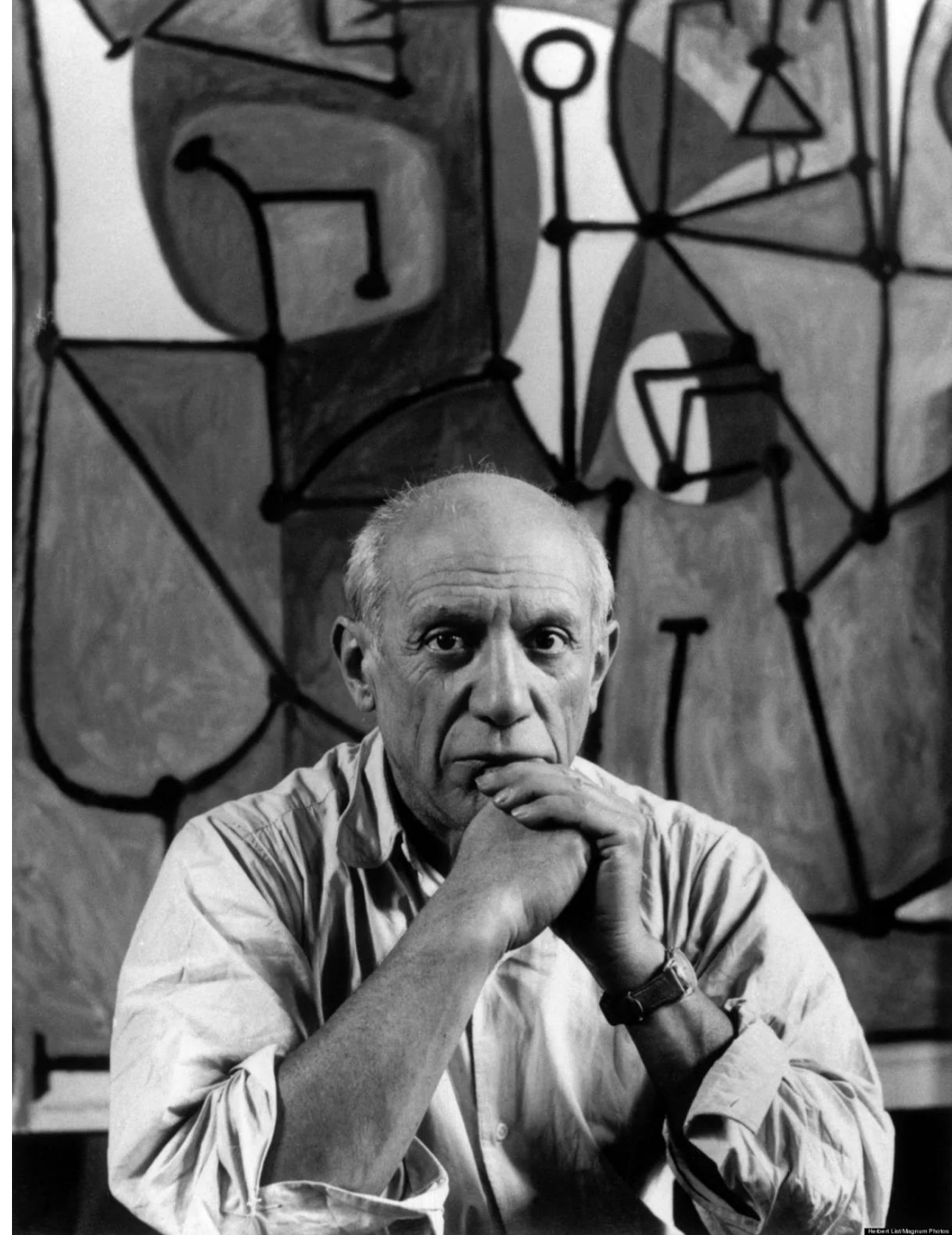
“THE BAD ARTISTS
IMITATE, THE GREAT
ARTISTS STEAL.”

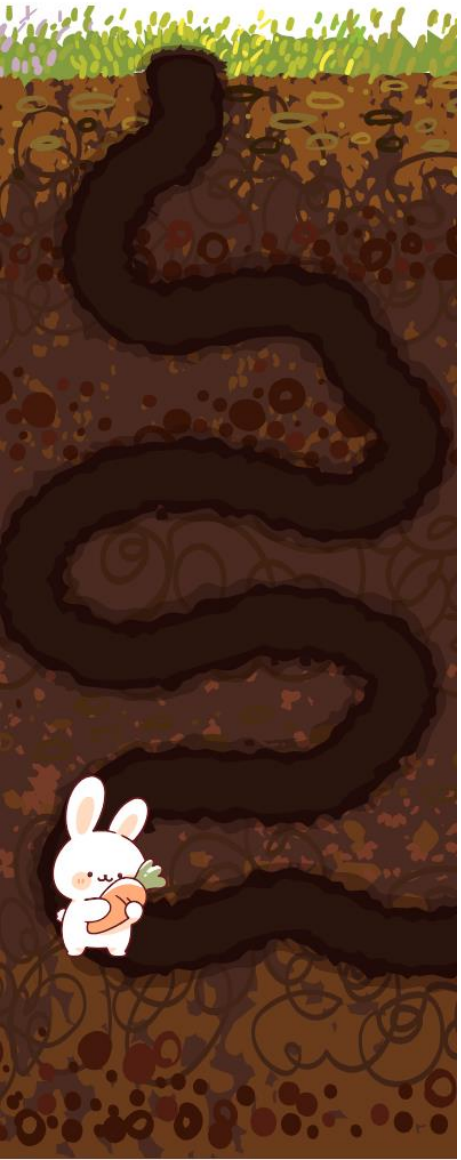
~~PABLO PICASSO~~
BANKSY

Rabbit Hole
Progress Tracking



“Good artists copy,
great artists steal.”
- *Pablo Picasso*





**“Immature poets imitate;
mature poets steal [...].**

The good poet welds his
theft into a whole of
feeling which is unique,
utterly different than that
from which it is torn.”

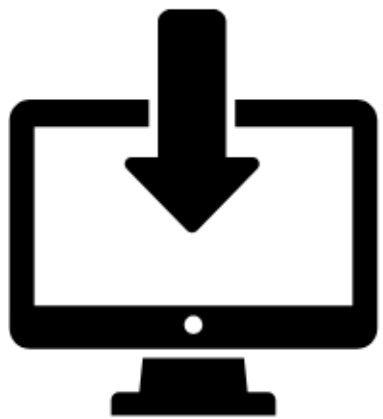
- T.S. Eliot



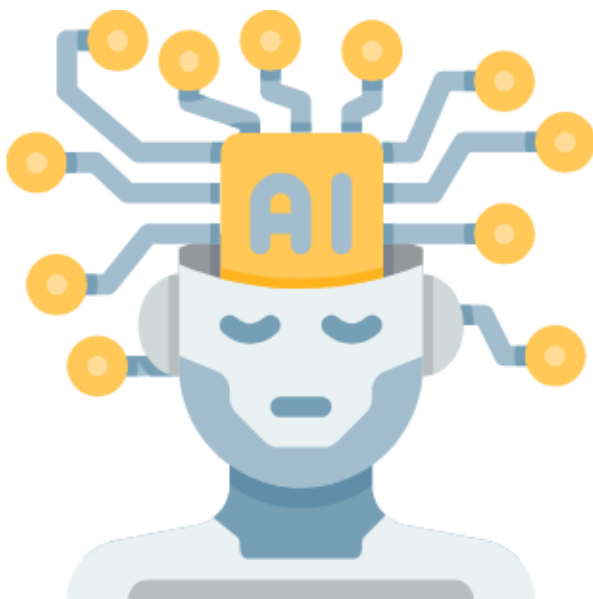
Rabbit Hole
Progress Tracking



Why?



Input



Generative Model



Output

Why? Philosophical Reason

Stefano Ermon

- Home
- CV
- Teaching
- Awards
- Publications
- GitHub
- Personal



Stefano Ermon [Follow @StefanoErmon](#)

Associate Professor
Department of Computer Science
Stanford University

Office: Gates Building #330
Phone: (650) 498-9942
Email: ermon AT cs.stanford.edu

About Me

I am an Associate Professor in the Department of Computer Science at Stanford University. My research is in machine learning and generative AI. I like to develop principled methods in

Teaching

- Winter 2022/2023: [Probabilistic Graphical Models \(CS 228\)](#)
- Winter 2021/2022: [Probabilistic Graphical Models \(CS 228\)](#)
- Fall 2021/2022: [Deep Generative Models \(CS 236\)](#)
- Fall 2021/2022: [Data for Sustainable Development \(CS 325B\)](#)
- Winter 2020/2021: [Probabilistic Graphical Models \(CS 228\)](#)
- Winter 2019/2020: [Probabilistic Graphical Models \(CS 228\)](#)
- Fall 2019/2020: [Deep Generative Models \(CS 236\)](#)

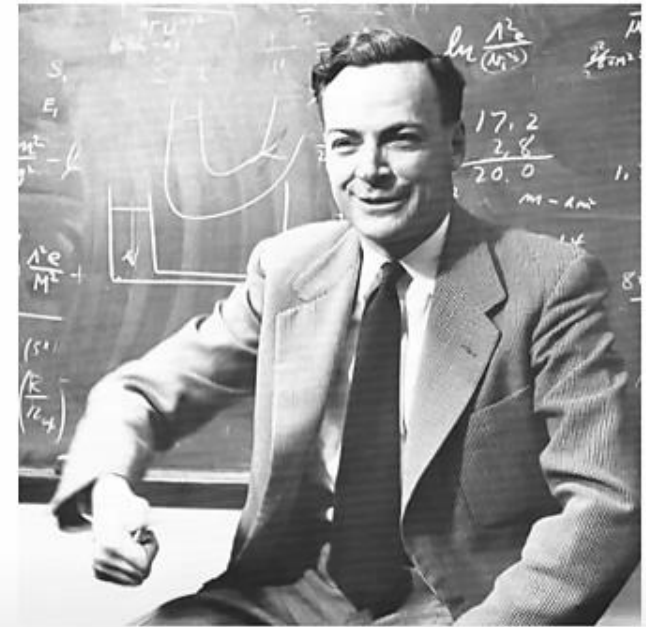
Deep Generative Models
Stanford - CS236

Why? Philosophical Reason

Introduction



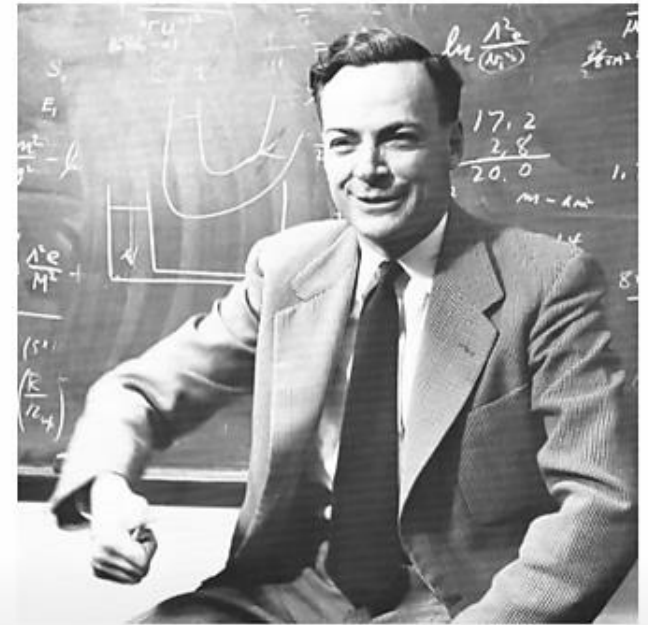
What I cannot create,
I do not understand.



Richard Feynman: “*What I cannot create, I do not understand*”

Why? Philosophical Reason

Introduction

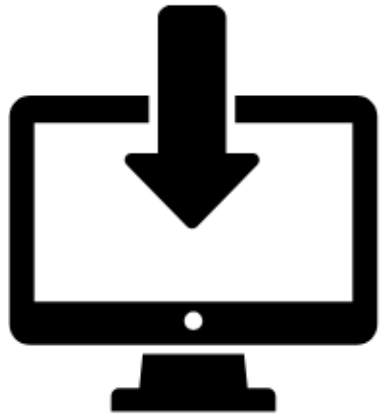


Richard Feynman: “*What I cannot create, I do not understand*”

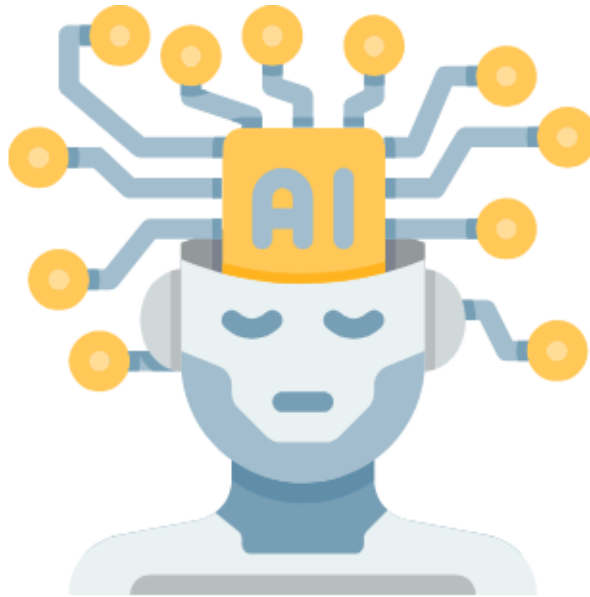
Generative modeling: “*What I understand, I can **create***”

Stanford

Why? Practical Applications



Input



Generative Model

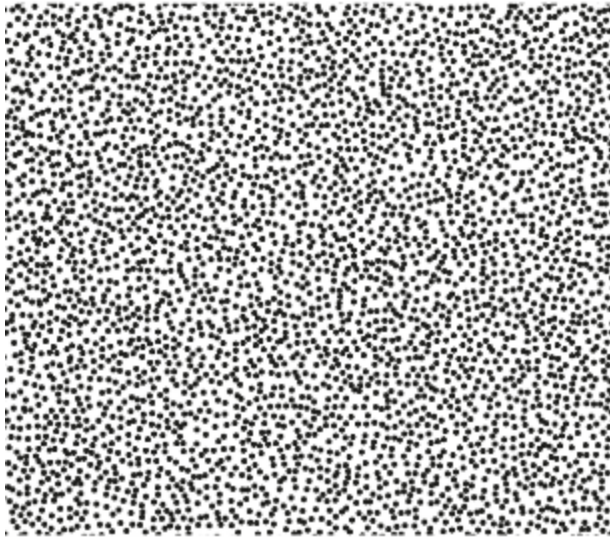


Output

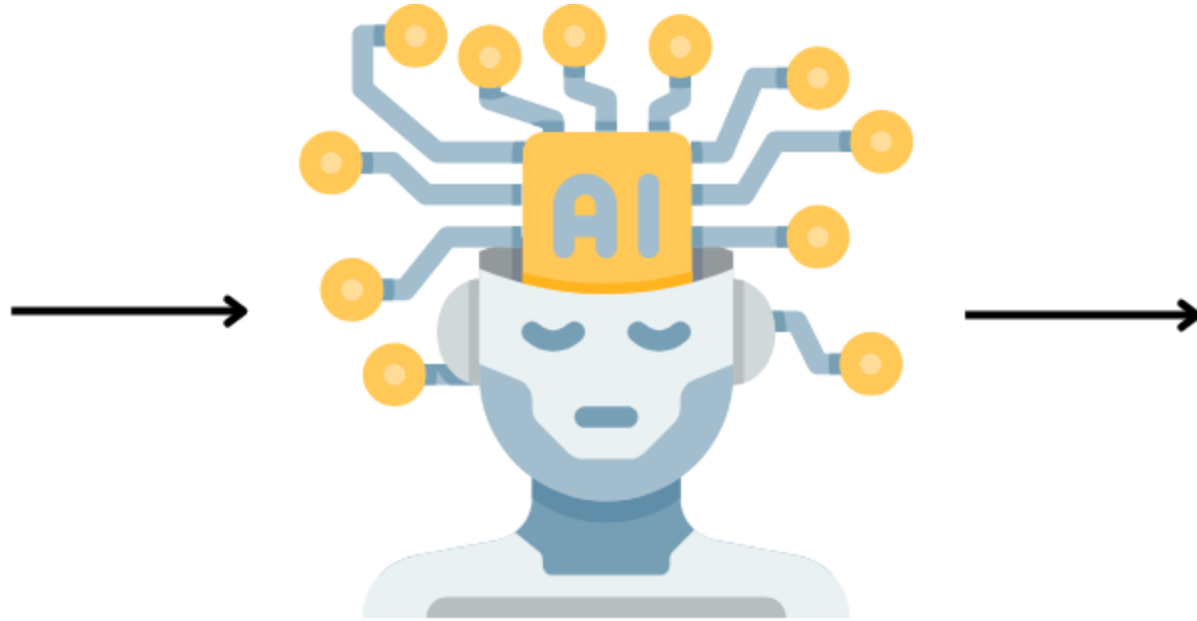
Why? Practical Applications

- Many problems can be formulated as image generation.
- Based on the input the approach can be:
 1. Noise-to-Image
 2. Text-to-Image
 3. Image-to-Image
 4. Text and Image-to-Image

Noise-to-Image Generation



Noise



Generative Model



Output

Noise-to-Image Generation

Example: <https://thispersondoesnotexist.com/>

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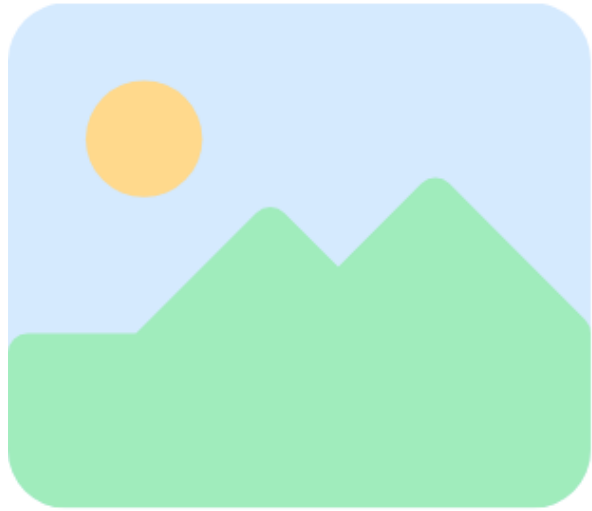
Enter the code

6718 8786

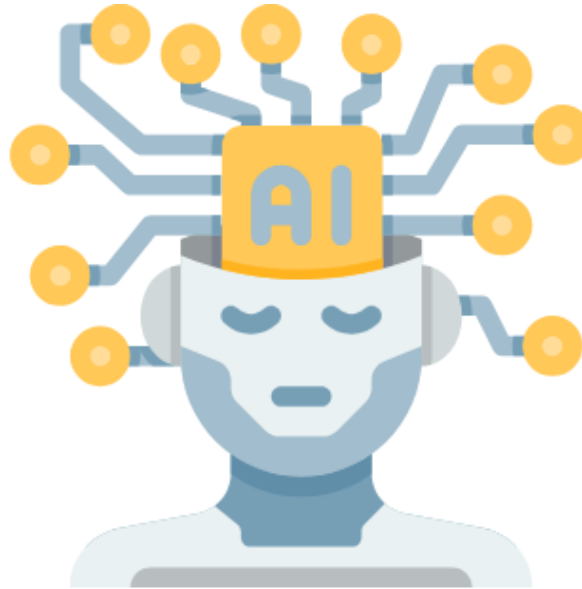


Or use QR code

Image-to-Image Generation



Image



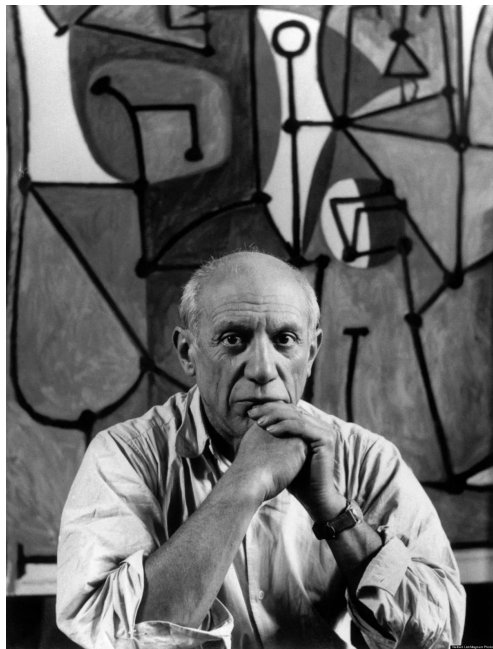
Generative Model



Output

Image-to-Image Generation

Example: Style Transfer - <https://reiinakano.com/arbitrary-image-stylization-tfjs/>



Input Image



Style Image

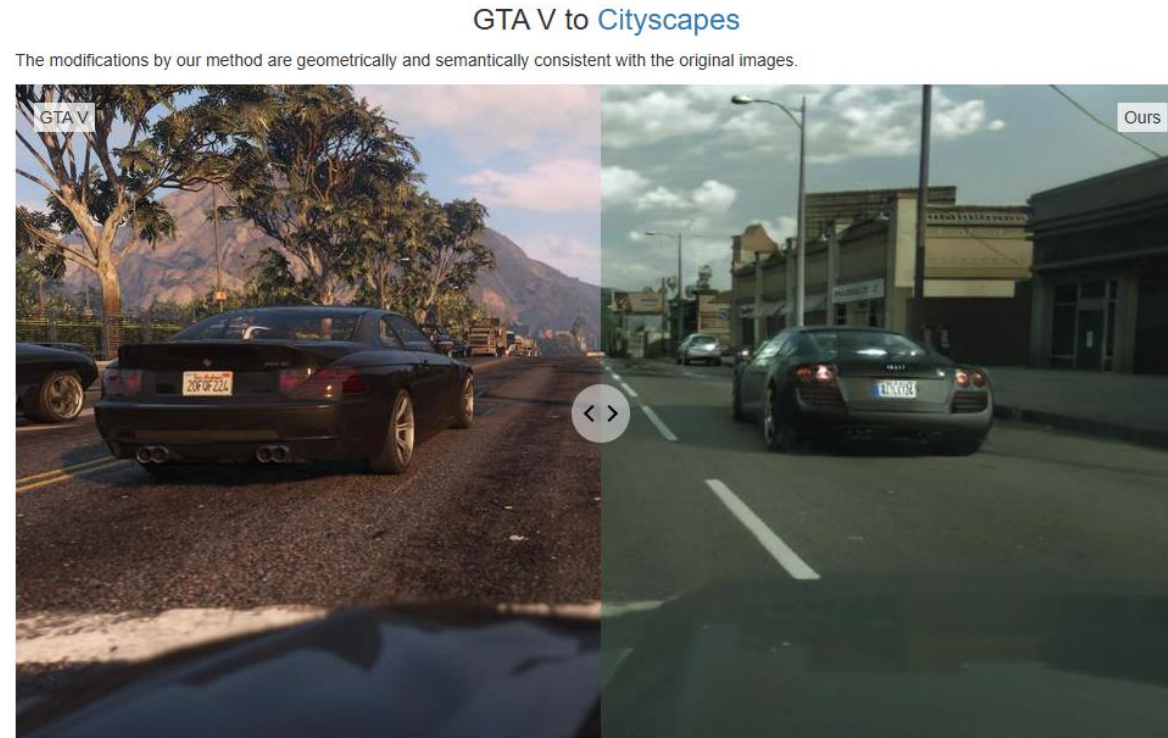


Result

Image-to-Image Generation

Application: Enhancing Photorealism Enhancement ^[1]

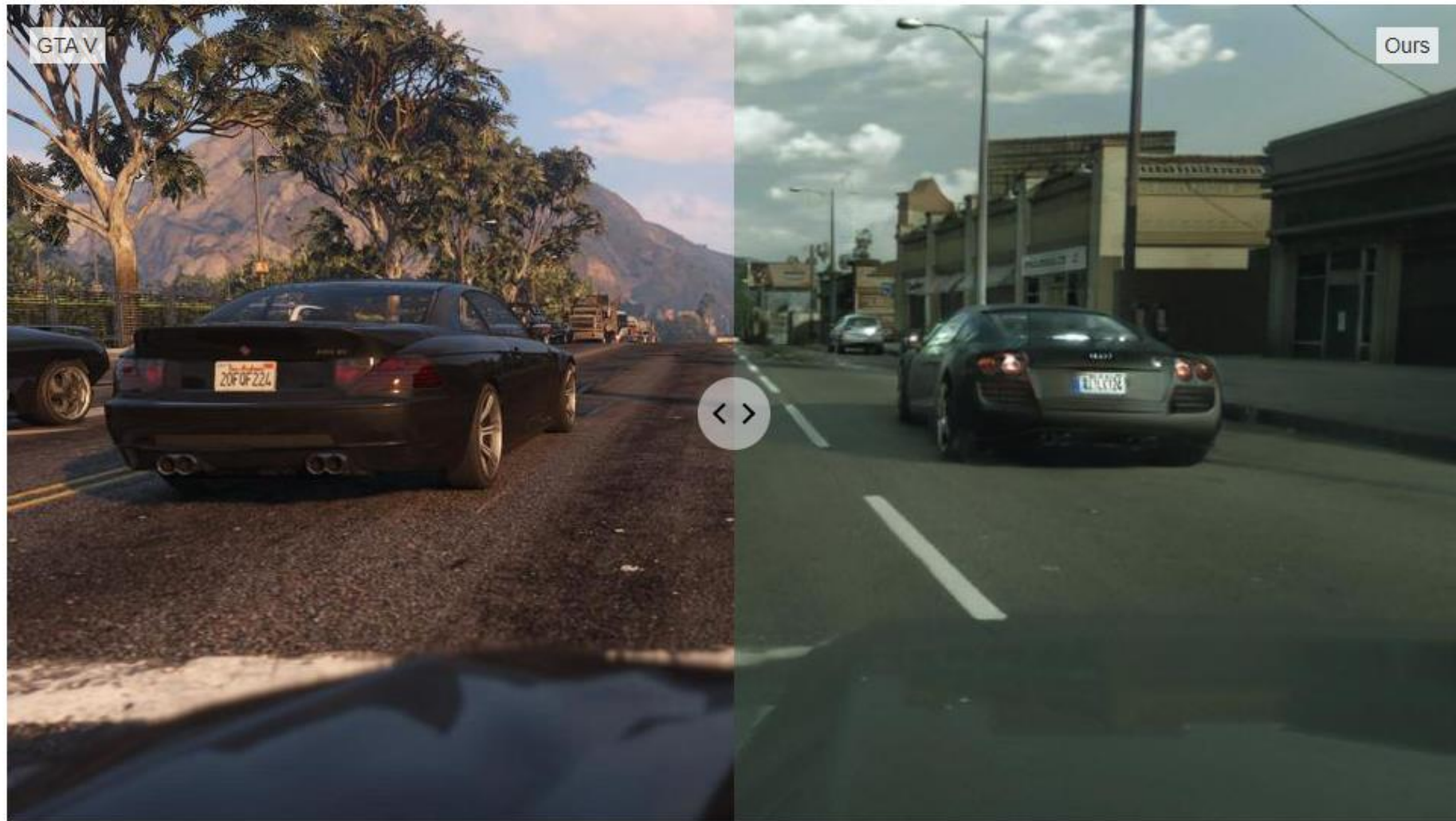
<https://isl-org.github.io/PhotorealismEnhancement/>



[1] Richter, S.R., AlHaija, H.A. and Koltun, V., 2022. Enhancing photorealism enhancement. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2), pp.1700-1715.

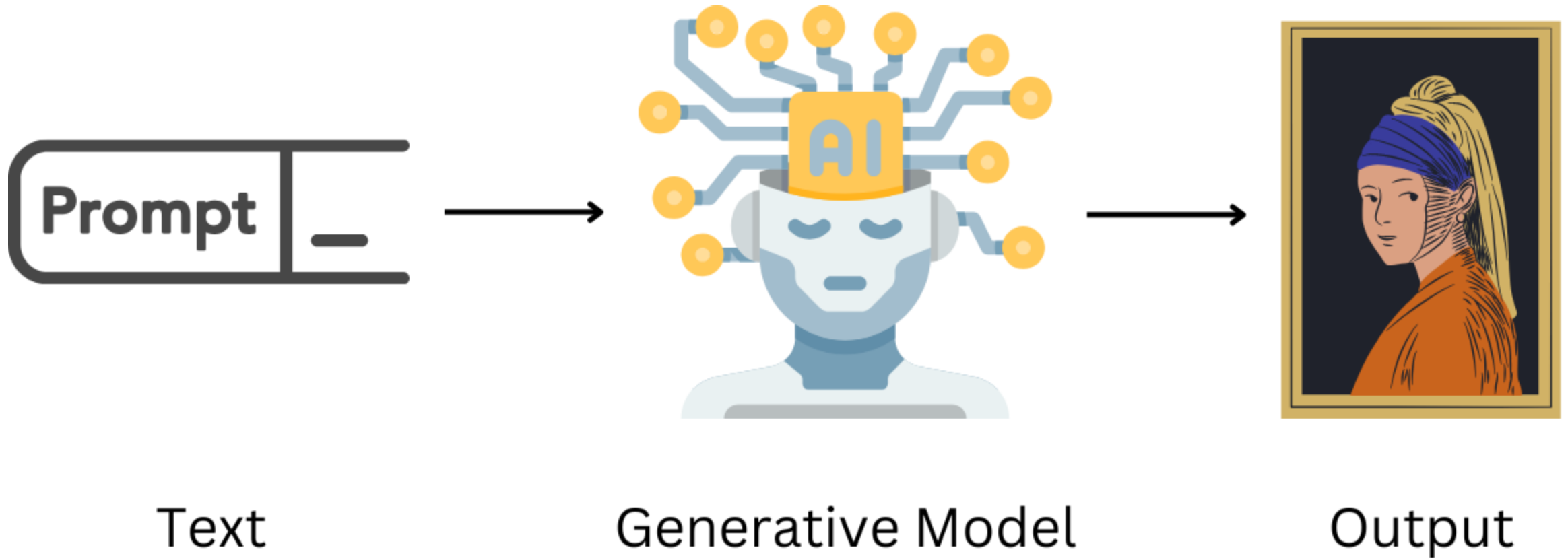
GTA V to Cityscapes

The modifications by our method are geometrically and semantically consistent with the original images.



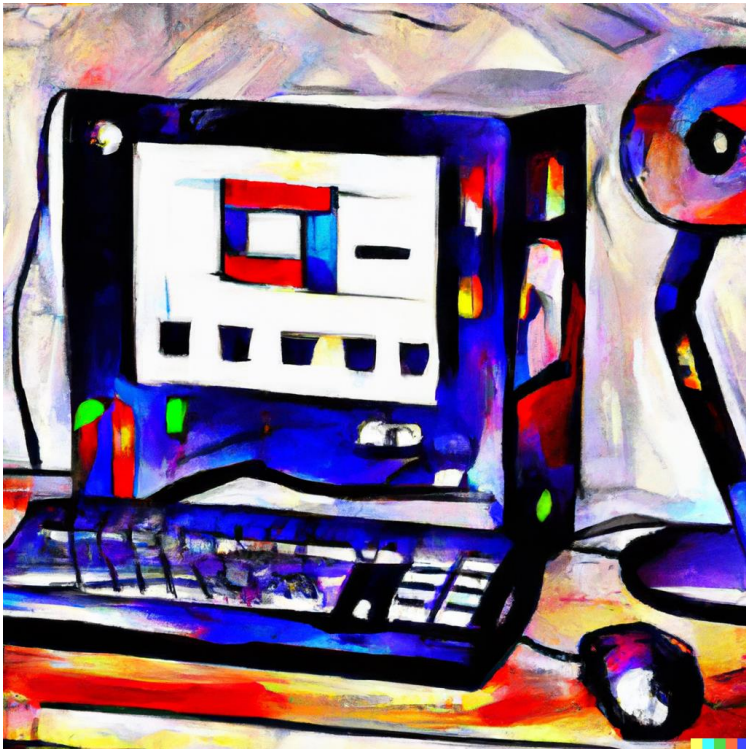
<https://isl-org.github.io/PhotorealismEnhancement/>

Text-to-Image Generation



Text-to-Image Generation

Example: DALL-E 2 <https://openai.com/index/dall-e-2/>



*Prompt: A painting of a computer
in the style of Pablo Picasso*

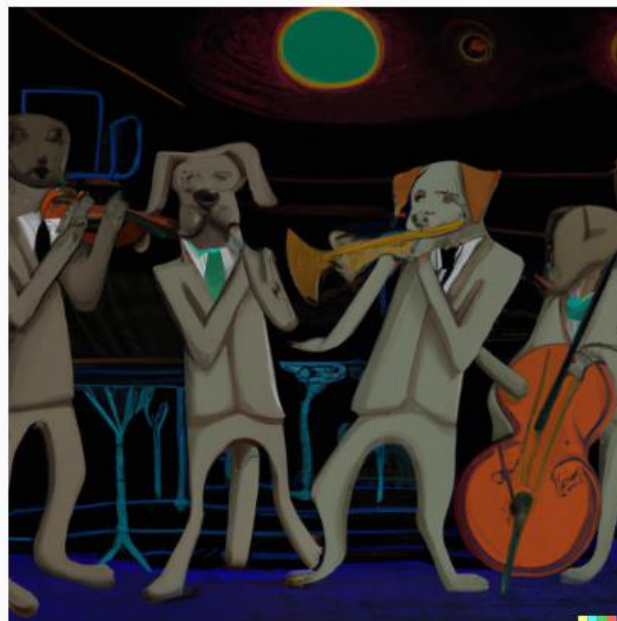
Text-to-Image Generation

Example: DALL-E 2 <https://openai.com/index/dall-e-2/>

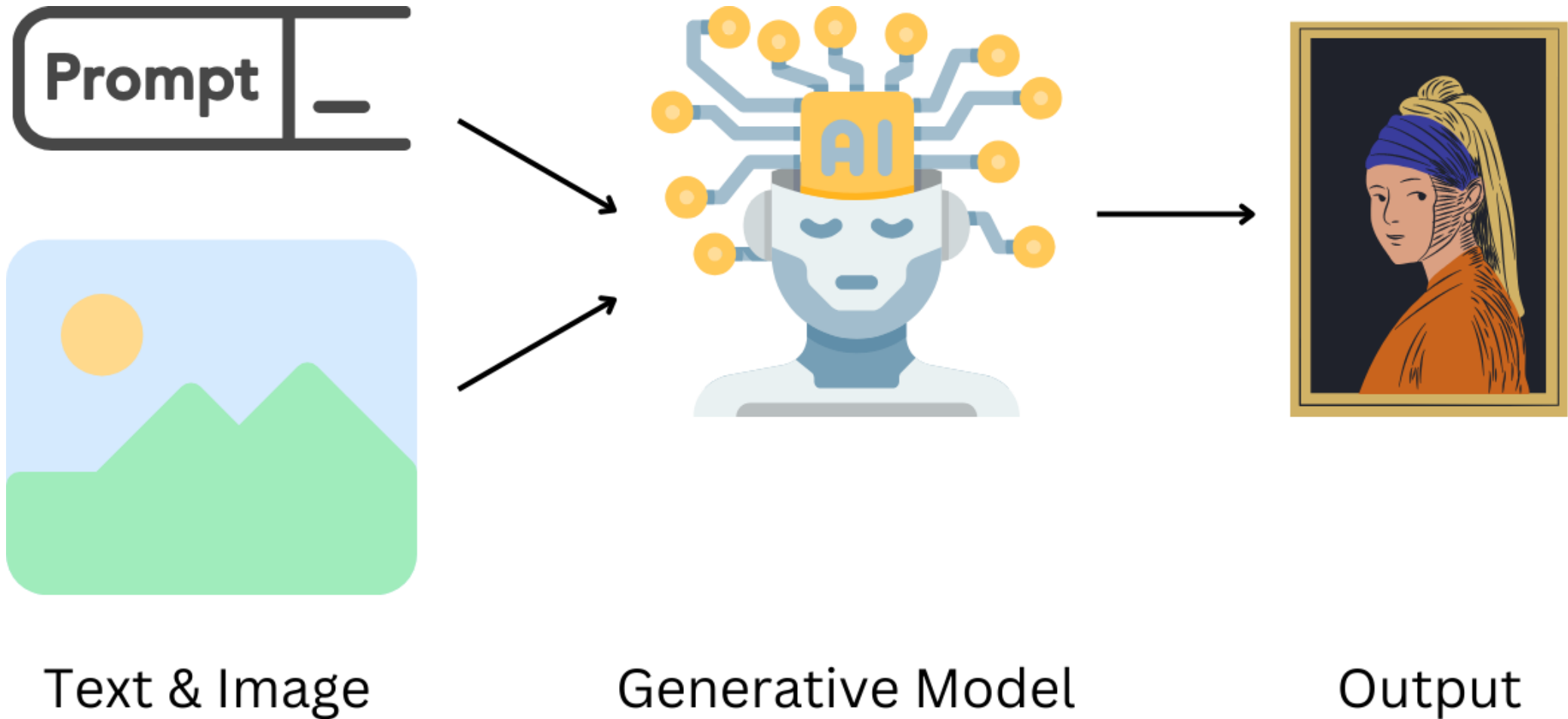
A hyper-realistic image of a dog quartet playing jazz instruments in a moody club during the night

Generate

A hyper-realistic image of a dog quartet playing jazz instruments in a moody club during the night



Multimodal Image Generation



Multimodal Image Generation

Example: ControlNet <https://stablediffusion-online.com/controlnet.php>



"A painting in the style of Pablo Picasso"

Prompt

A painting in the style of Pablo Picasso

Input



Result

Why? Practical Applications

- Applications:

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Why? Practical Reason

- Applications:
 1. Image Generation for Dataset Extension
 2. Creativity Boosting
 3. Image Inpainting

Inpainting



Application examples of inpainting techniques: photo restoration (top left: image from (Bertalmio et al., [2000](#))), text removal (top right: image from (Bertalmio et al., [2000](#))), undesired target removal (bottom left: image from (Chen, [2018](#))), and face verification (bottom right: image from (Zhang et al., [2018c](#)))

Inpainting



[3] Corneanu, Ciprian, Raghudeep Gadde, and Aleix M. Martinez. "Latentpaint: Image inpainting in latent space with diffusion models." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. 2024.

Why? Practical Reason

- Applications:
 1. Image Generation for Dataset Extension
 2. Creativity Boosting
 3. Image Inpainting
 4. Image Colorization

Image Colorization

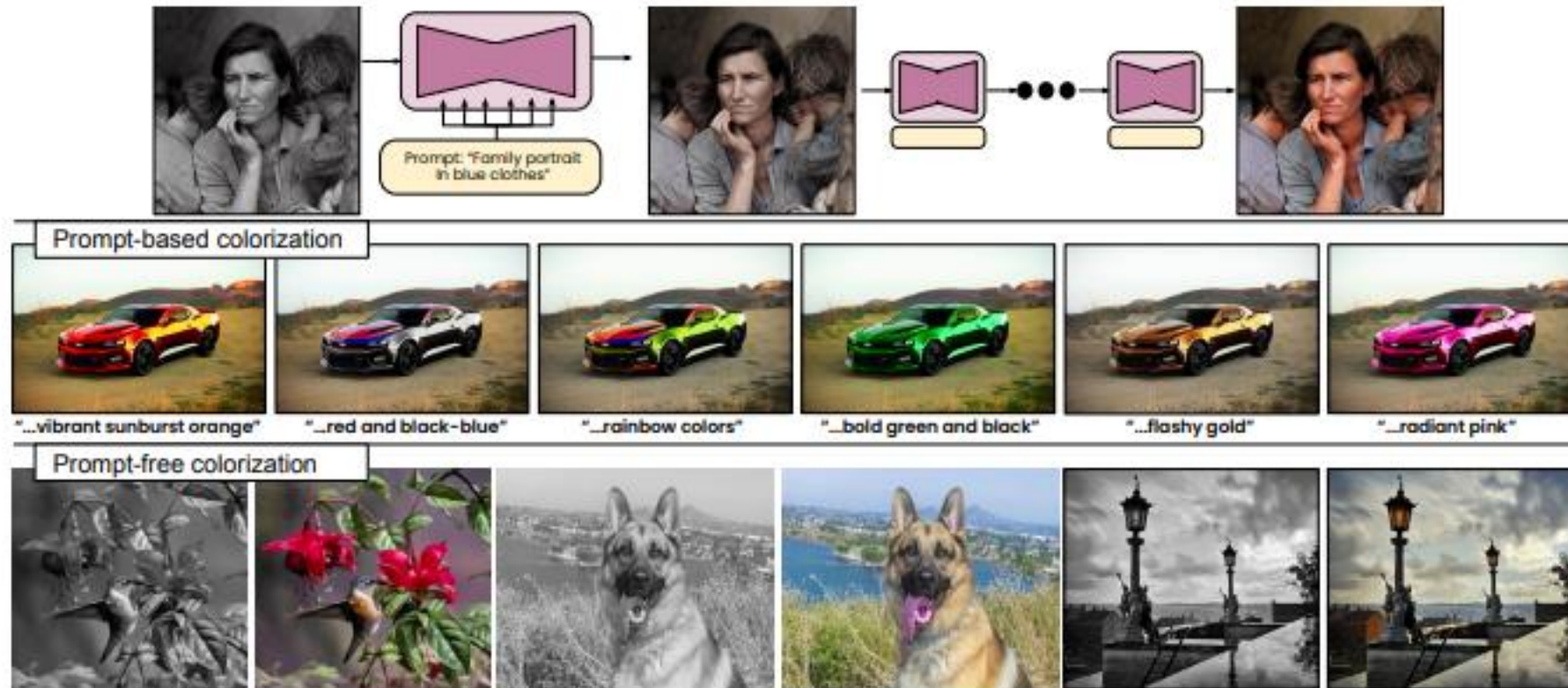


Image Colorization

- Interesting Git Repo: <https://github.com/ErwannMillon/Color-diffusion>

Why? Practical Reason

- Applications:
 1. Image Generation for Dataset Extension
 2. Creativity Boosting
 3. Image Inpainting
 4. Image Colorization
 5. Image Super-Resolution

Image Super-Resolution



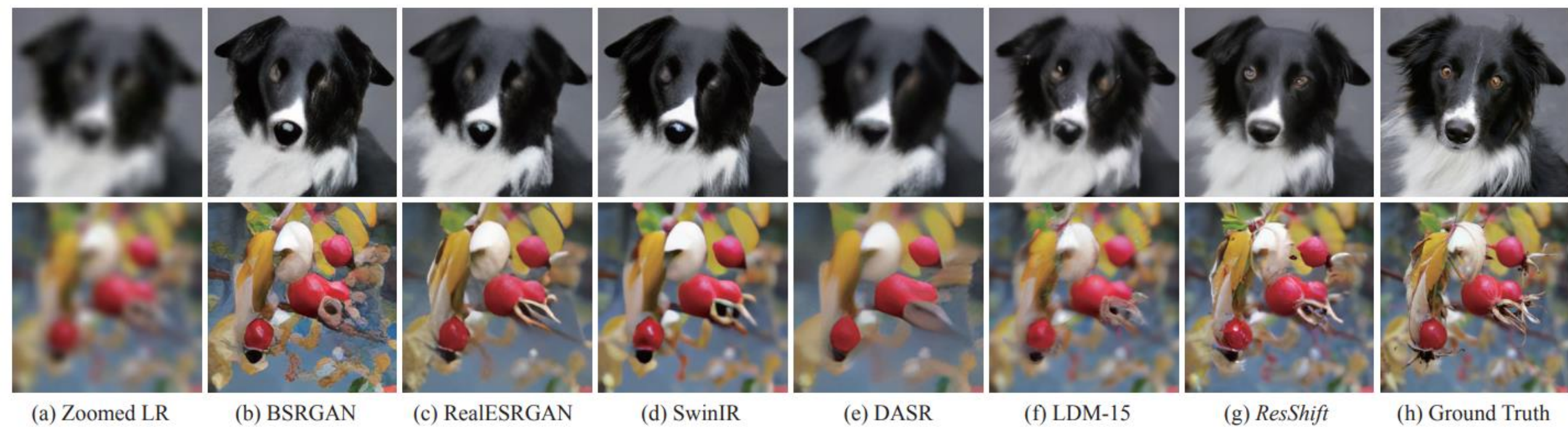
original



?upscale=true

Image from <https://www.imgix.com/blog/ai-powered-image-super-resolution>

Image Super-Resolution

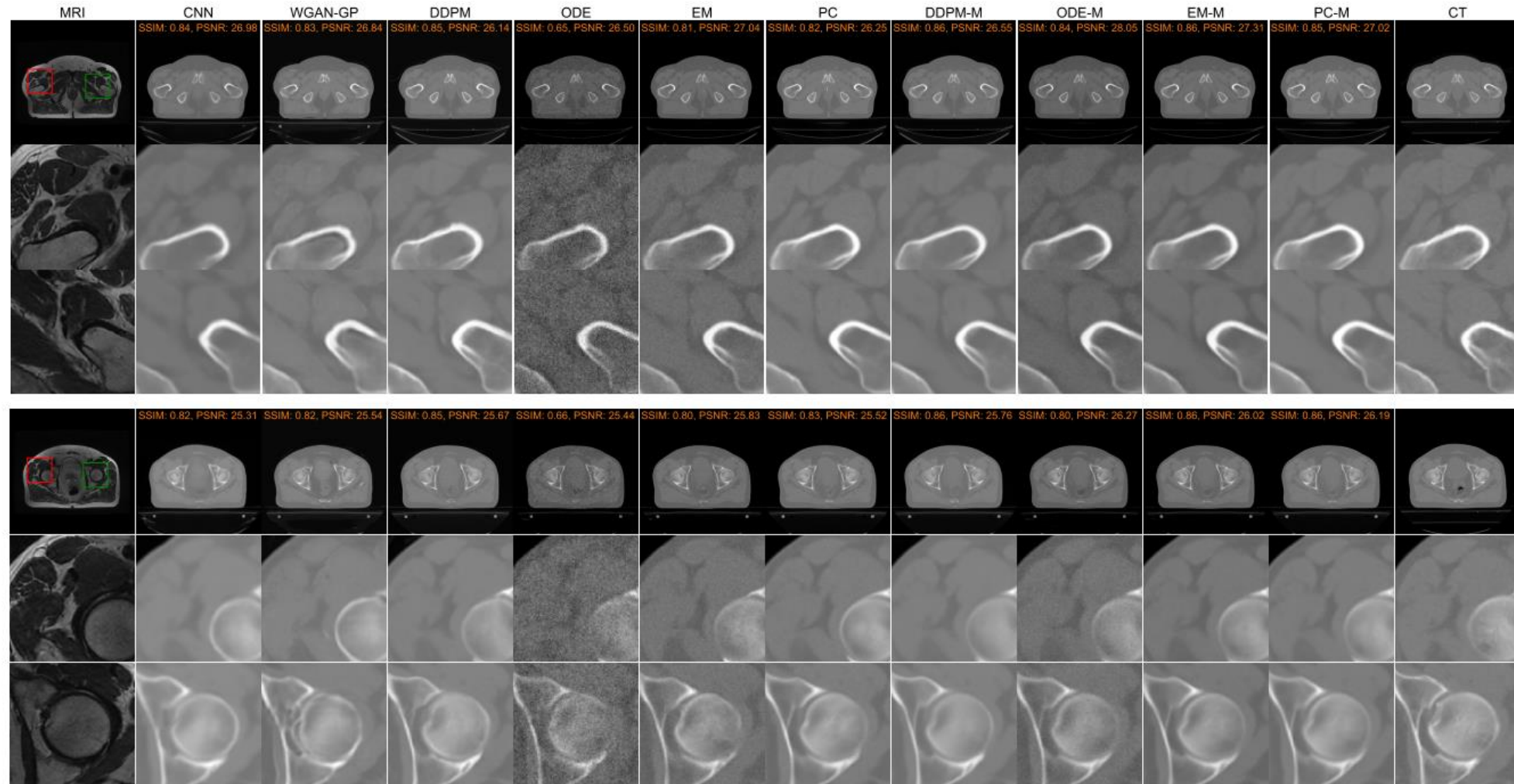


[5] Yue, Zongsheng, Jianyi Wang, and Chen Change Loy. "Resshift: Efficient diffusion model for image super-resolution by residual shifting." *Advances in Neural Information Processing Systems* 36 (2024).

Why? Practical Reason

- Some Applications:
 1. Image Generation for Dataset Extension
 2. Creativity Boosting
 3. Image Inpainting
 4. Image Colorization
 5. Image Super-Resolution
 6. Medical Image Conversion

Transform MRI into CT



[6] Lyu, Qing, and Ge Wang. "Conversion between CT and MRI images using diffusion and score-matching models." arXiv preprint arXiv:2209.12104 (2022).

Why Stable Diffusion?

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Why Stable Diffusion?

- Generative Models:
 1. Variational Autoencoders (2013) ^[7]
 2. Generative Adversarial Networks (2014) ^[8]
 3. Stable Diffusion (2020) ^[9]

[7] Kingma, Diederik P. "Auto-encoding variational bayes." arXiv preprint arXiv:1312.6114 (2013).

[8] Goodfellow, Ian, et al. "Generative adversarial nets." Advances in neural information processing systems 27 (2014).

[9] Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." Advances in neural information processing systems 33 (2020): 6840-6851.

Why Stable Diffusion?

- Generative Models:

1. Variational Autoencoders (2013) ^[7]
2. Generative Adversarial Networks (2014) ^[8]
3. Denoising Diffusion Probabilistic Models (DDPMs) (2020) ^[9]
4. Deep Unsupervised Learning using Nonequilibrium Thermodynamics (2015) ^[10]

[7] Kingma, Diederik P. "Auto-encoding variational bayes." arXiv preprint arXiv:1312.6114 (2013).

[8] Goodfellow, Ian, et al. "Generative adversarial nets." Advances in neural information processing systems 27 (2014).

[9] Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." Advances in neural information processing systems 33 (2020): 6840-6851.

[10] Sohl-Dickstein, Jascha, et al. "Deep unsupervised learning using nonequilibrium thermodynamics." International conference on machine learning. PMLR, 2015.

Why Stable Diffusion?

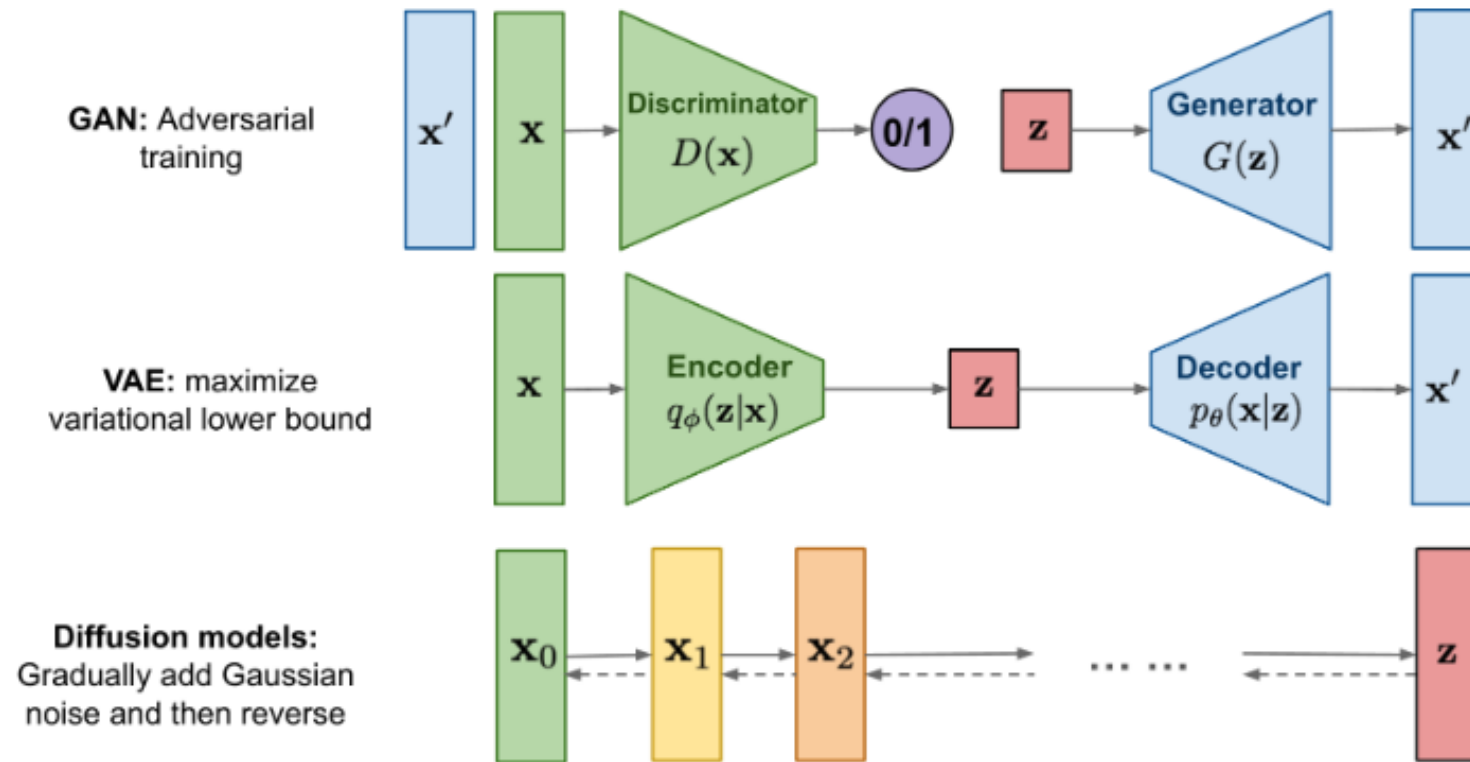


Image from <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

Intuition Check

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Why Stable Diffusion?

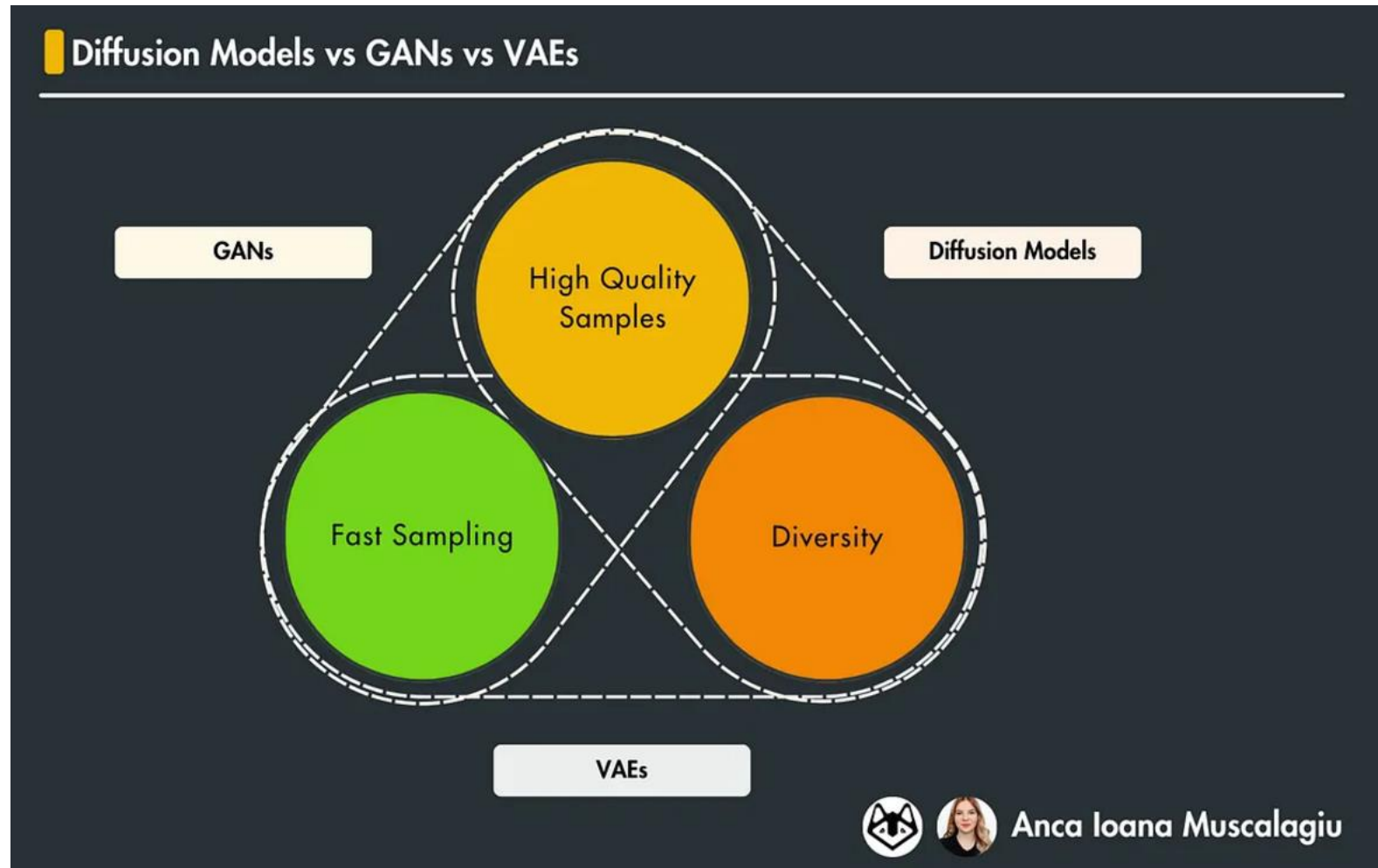


Image from <https://medium.com/decodingml/the-abcs-of-diffusion-models-51902a331068>

Why Stable Diffusion?

- High-quality samples
- Diverse sample
- **Cost:** Slow Sampling

What is diffusion?

 Despre noi ▾ Implică-te ▾ Resurse ▾

🌐 🌙 👤 Anonim ▾

× 🔍 caută [opțiuni](#)

 Forma **difuzie** este o variantă a lui **difuziune**.

sinteză (1)

definiții (17)

declinări

📘



Aceste definiții sunt compilate de echipa dexonline. Definițiile originale se află pe fila [definiții](#).
Puteți reordona filele pe pagina de [preferințe](#).

ascunde

arată:

☒ sensuri secundare

☒ expresii

☐ exemple

☐ surse

difuziune, difuziuni / **difuzie**, difuzii substantiv feminin

1. Împrăștierea în toate direcțiile

razelor unui fascicul de lumină, a undelor de radio etc. care trec printr-un mediu translucid sau care se reflectă când întâlnesc o suprafață cu asperități.

sinonime: propagare răspândire împrăștiere

1.1. Difuzare de unde radiofonice.
sinonime: difuzare

2. Pătrunderea moleculelor unui corp în masa altui corp cu care vin în contact.

2.1. Întrepătrundere a două lichide puse în contact fără a fi agitație.

etimologie:

limba franceză

diffusion

limba latină

diffusio, -onis

What is diffusion?

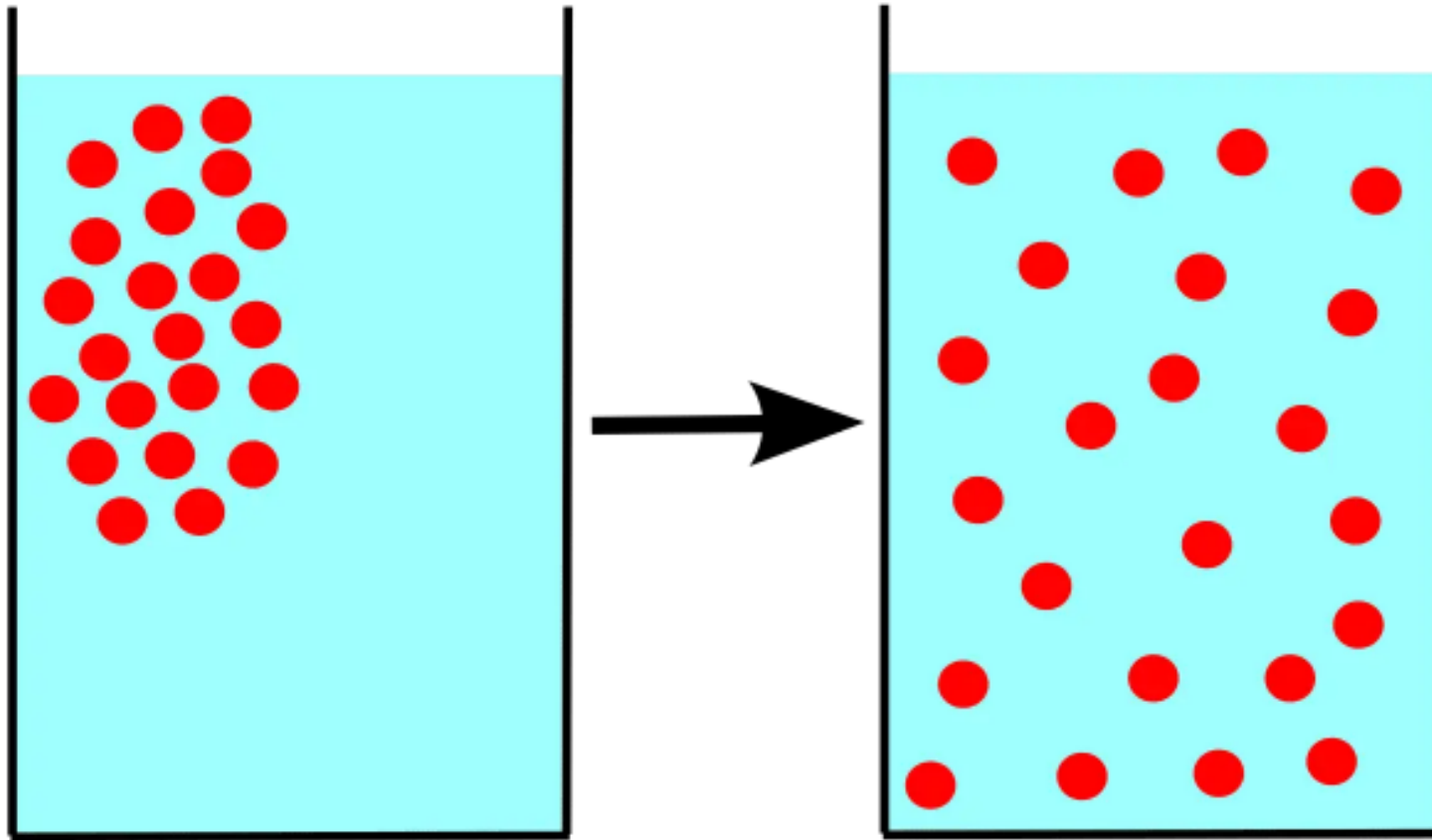


Image from <https://biologyjunction.com/osmosis-and-diffusion/>

What is diffusion?

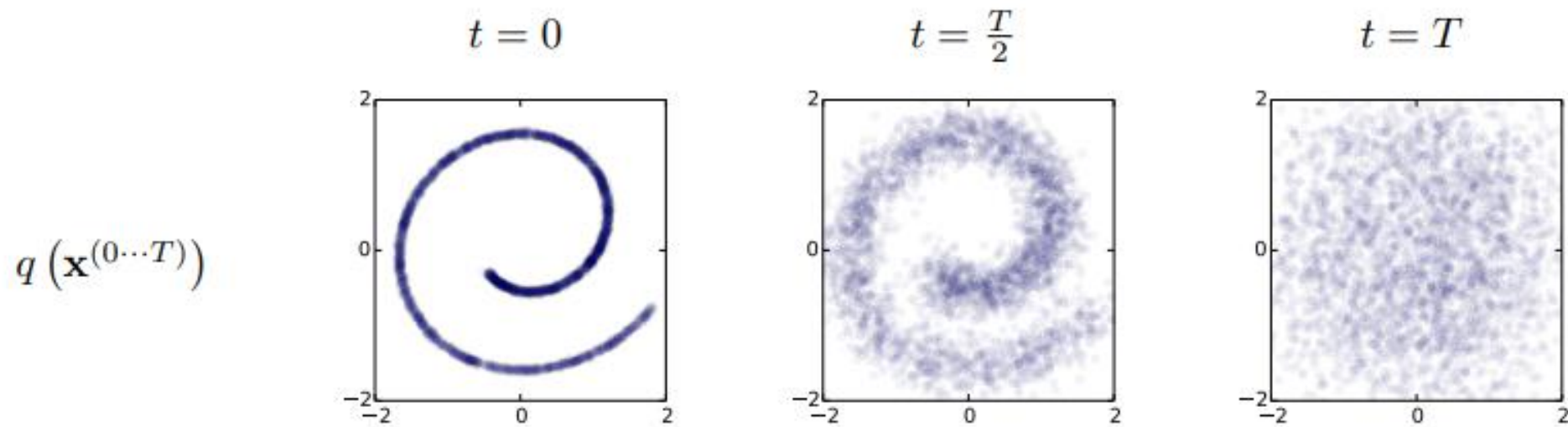


Image from Sohl-Dickstein, Jascha, et al. "Deep unsupervised learning using nonequilibrium thermodynamics." International conference on machine learning. PMLR, 2015.

What is diffusion?

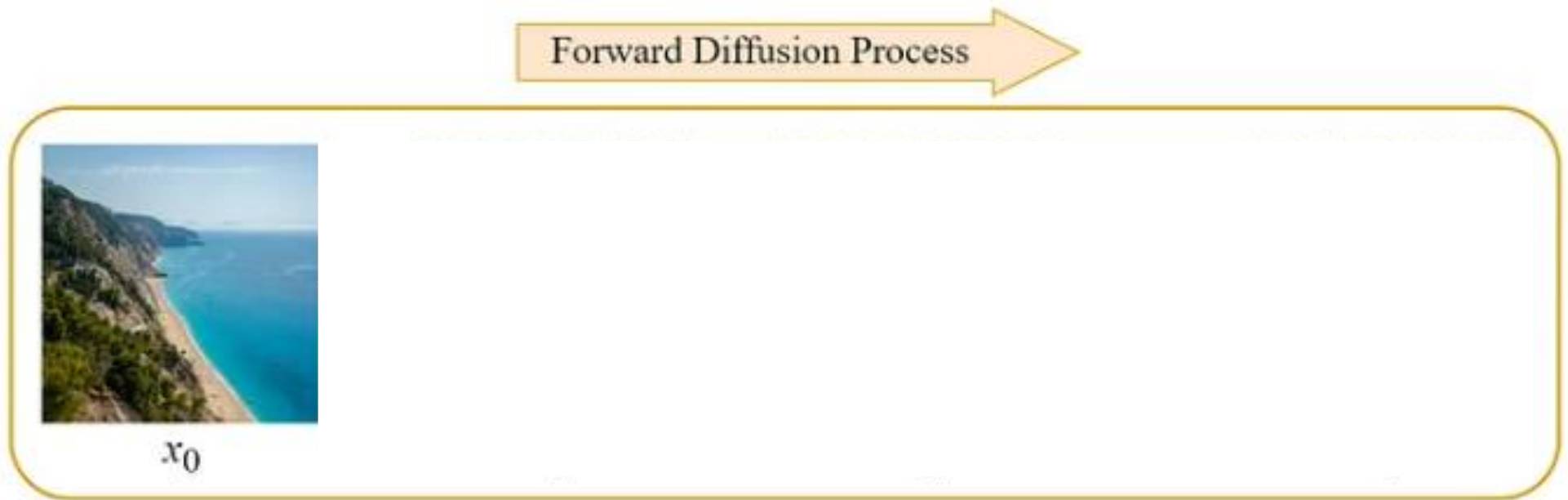


Image from <https://blog.marvik.ai/>

How does it work?

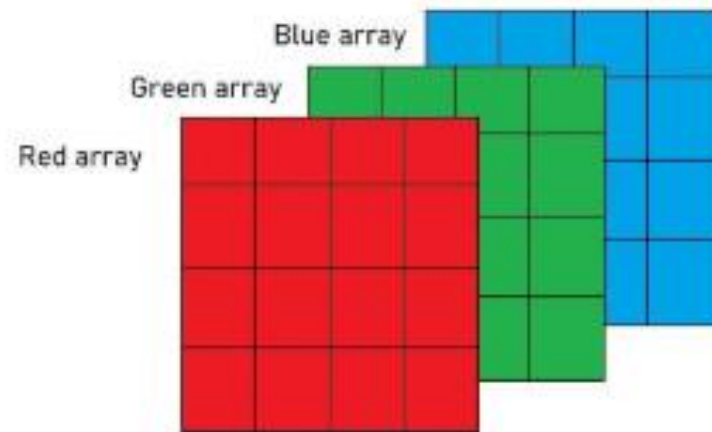
1. Forward Diffusion
2. “Reparameterization Trick”
3. Training Procedure
4. Generation Procedure

Forward Diffusion



Diffusion Markov Chain. Image from <https://medium.com/@jaskaranbhatia/summarizing-the-evolution-of-diffusion-models-insights-from-three-research-papers-6889339eba4>

Refresher – What is an image?



Arrays stacked over each other
to form a Digital Image.

Image from <https://www.geeksforgeeks.org/matlab-rgb-image-representation/>

Refresher – What is an image?

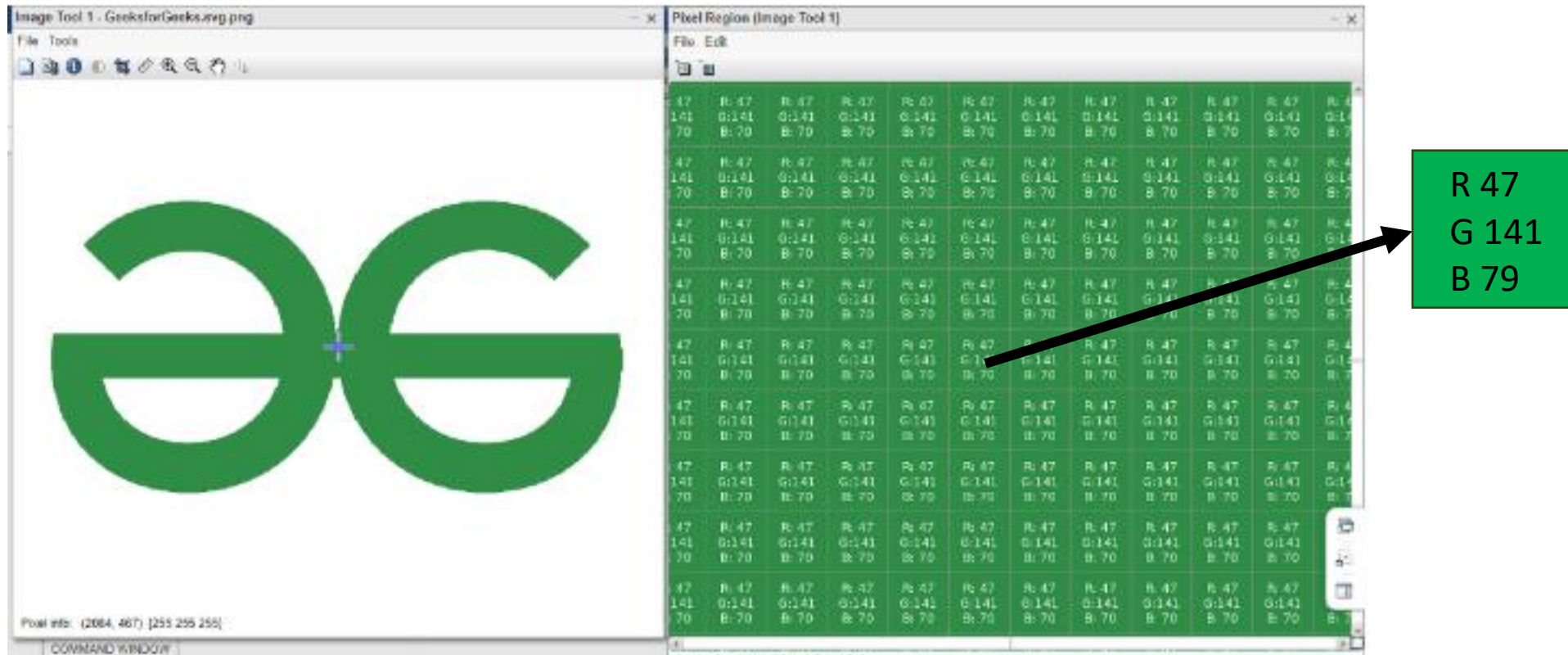


Image from <https://www.geeksforgeeks.org/matlab-rgb-image-representation/>

Forward Diffusion – Adding noise

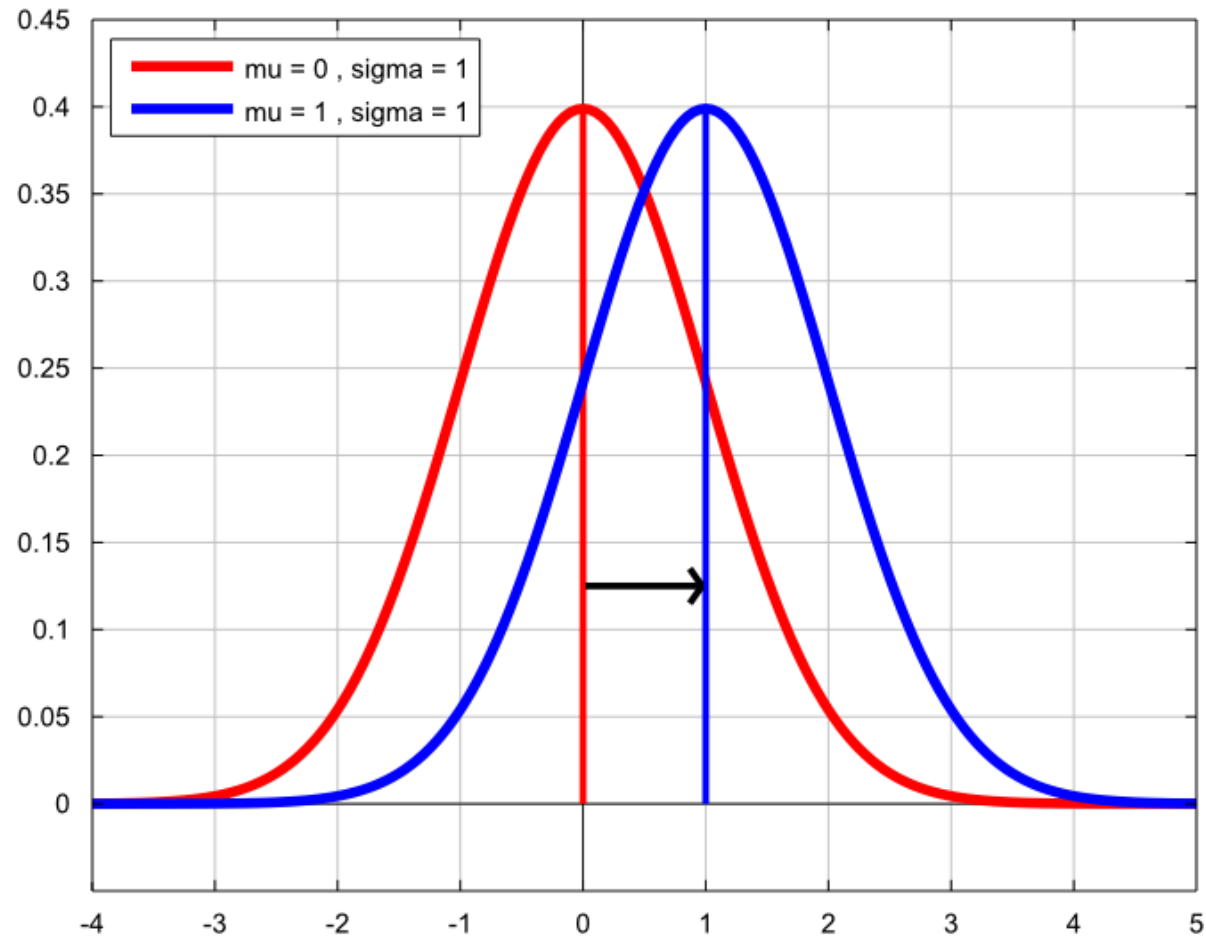


Image from <https://www.statlect.com/probability-distributions/normal-distribution>

Forward Diffusion – Adding noise

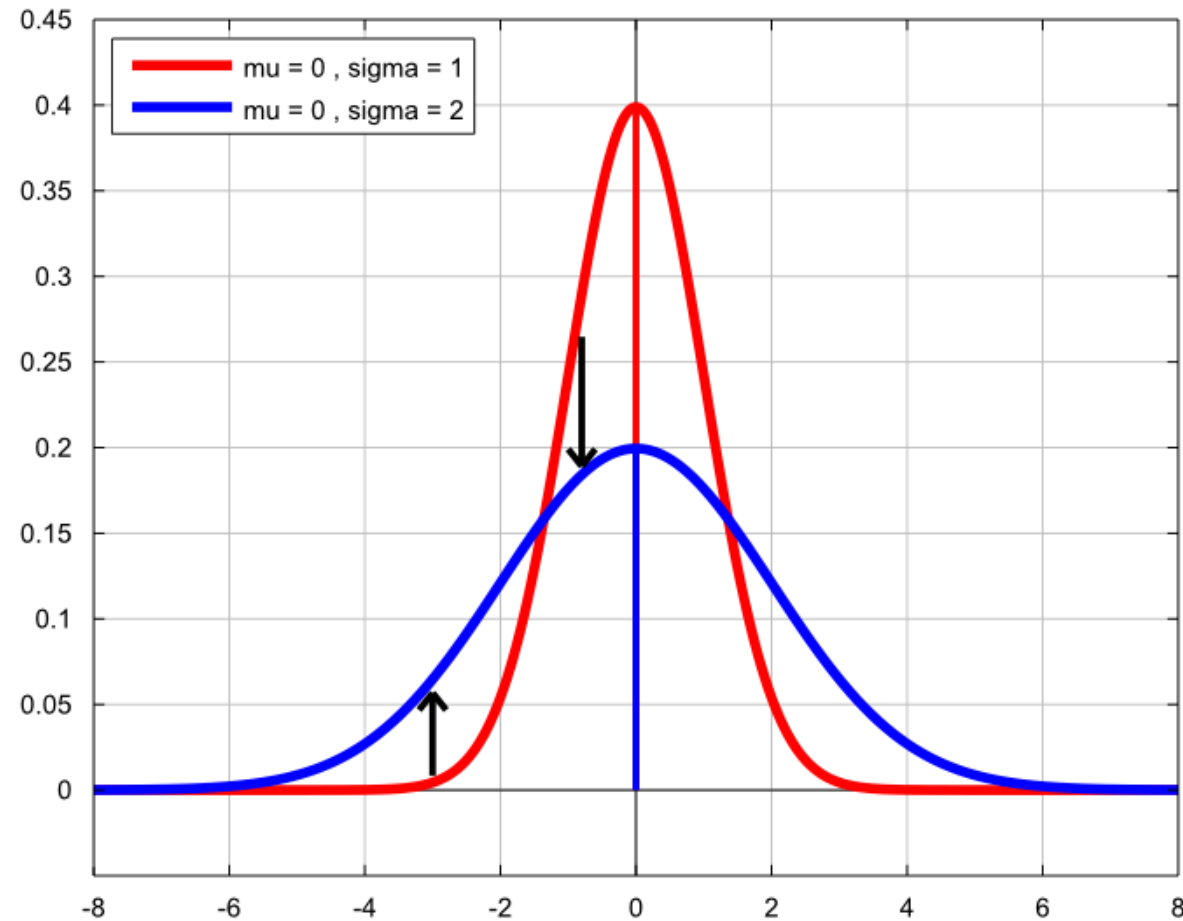


Image from <https://www.statlect.com/probability-distributions/normal-distribution>

Forward Diffusion – Adding noise

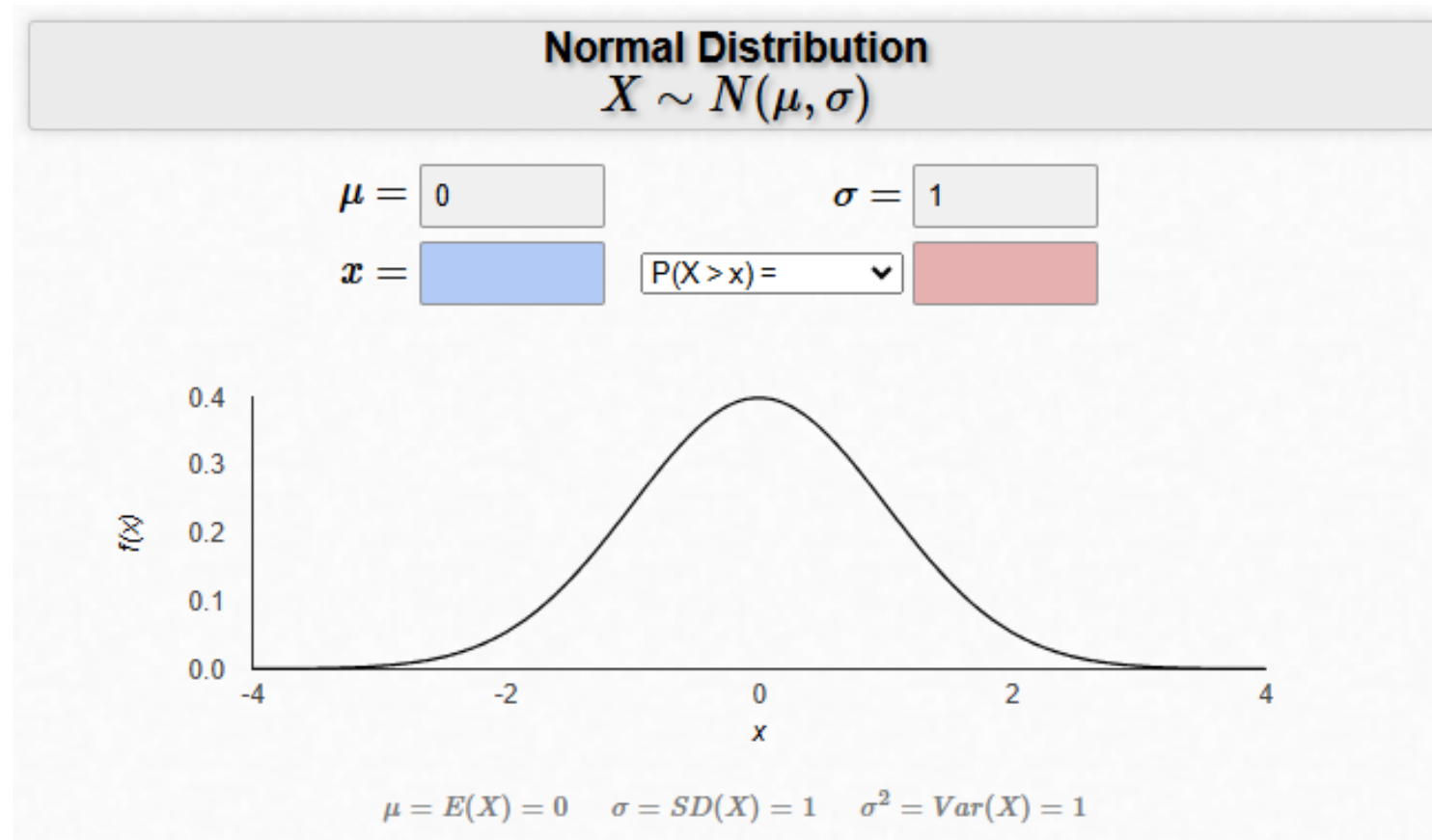


Image from <https://homepage.divms.uiowa.edu/~mbognar/applets/normal.html>

Forward Diffusion – Adding noise

How does the variance of the Gaussian noise influence the image?

<https://explore.albumentations.ai/transform/GaussNoise>

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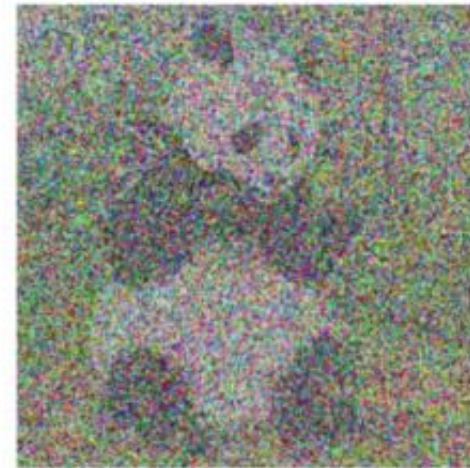
Forward Diffusion – Adding noise



(d)
Gaussian Noise
Mean=0, Variance = 0.002



(e)
Gaussian Noise
Mean=0, Variance = 0.05

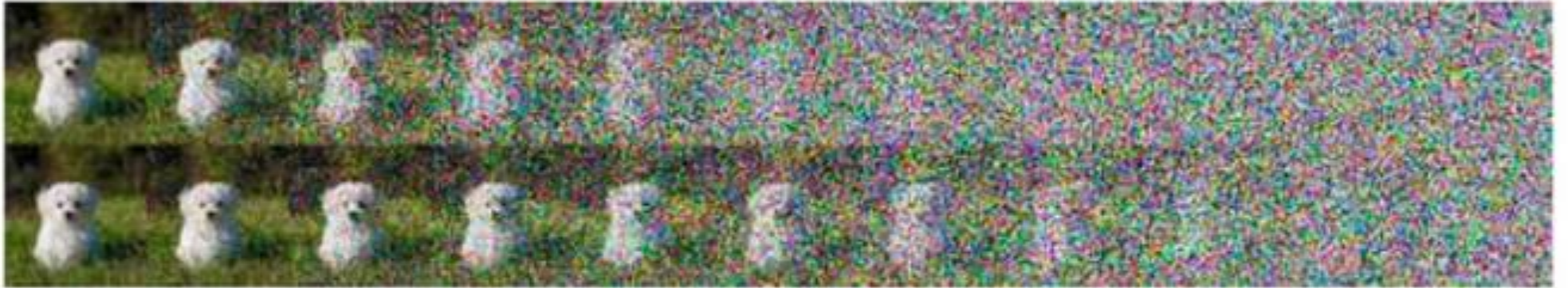


(f)
Gaussian Noise
Mean=0, Variance = 0.3

Image from https://www.researchgate.net/figure/Gaussian-Noise-with-different-variances-and-corresponding-decrypted-images_fig2_326215771

Forward Diffusion - Scheduling

Linear



Cosine



Image from Nichol, Alexander Quinn, and Prafulla Dhariwal. "Improved denoising diffusion probabilistic models." International conference on machine learning. PMLR, 2021.

Forward Diffusion



Image from <https://www.iguazio.com/glossary/diffusion-models/>

Reparameterization Trick



Image from <https://www.iguazio.com/glossary/diffusion-models/>

The Task: Reverse Diffusion

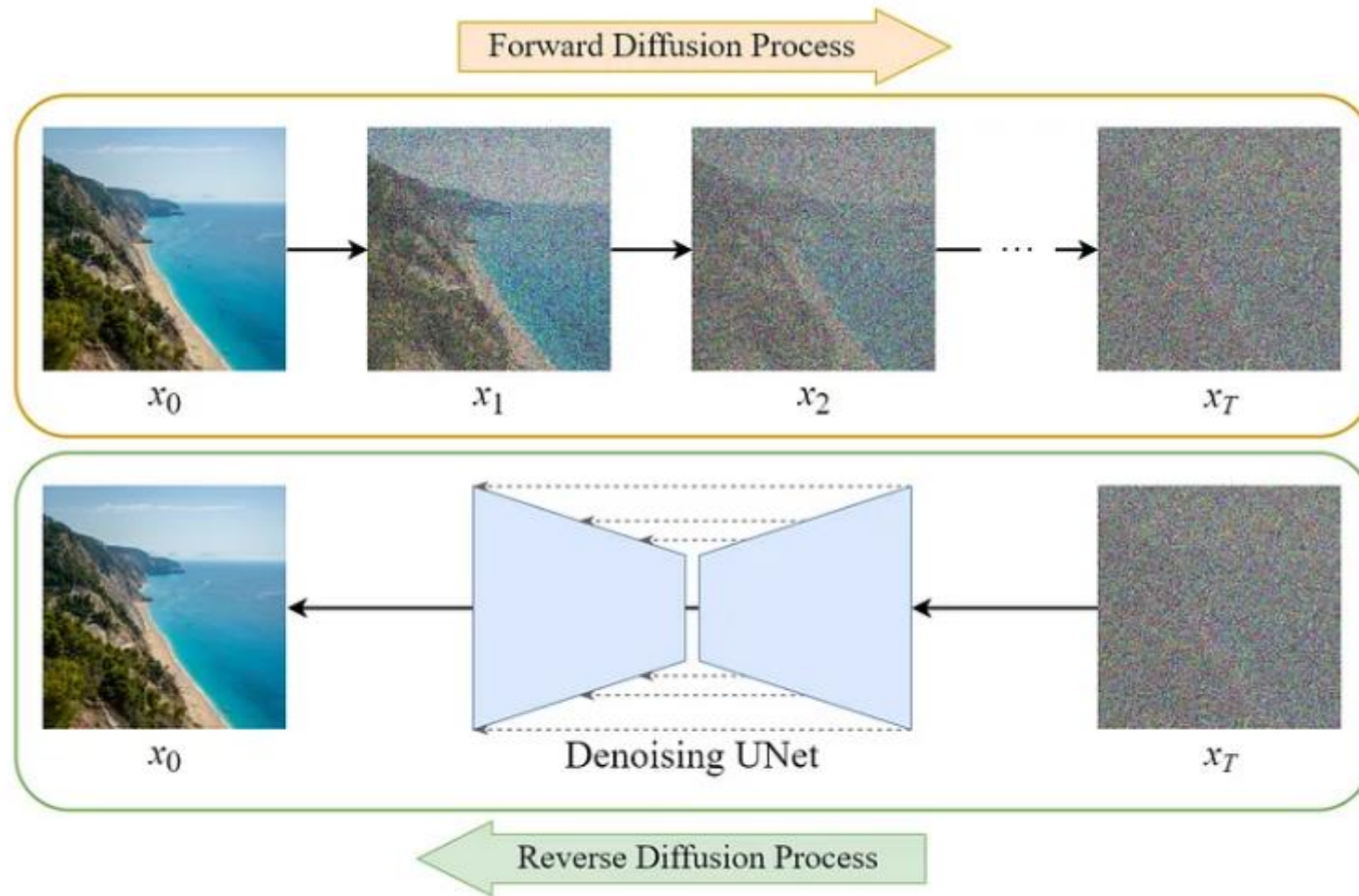


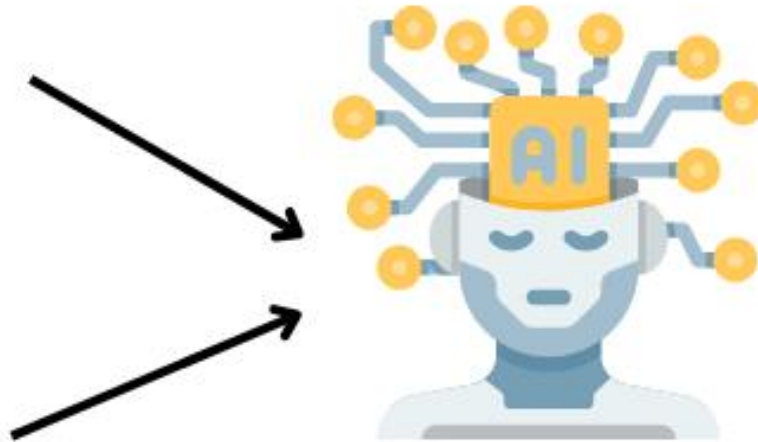
Image from <https://medium.com/@jaskaranbhatia/summarizing-the-evolution-of-diffusion-models-insights-from-three-research-papers-6889339eba4>

Training Procedure

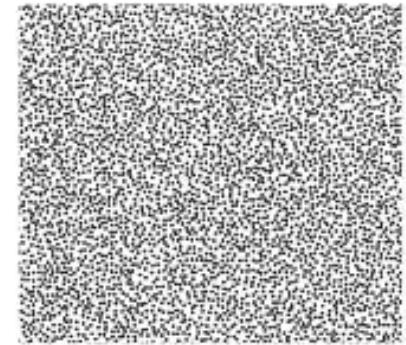


Image X_t

Time t

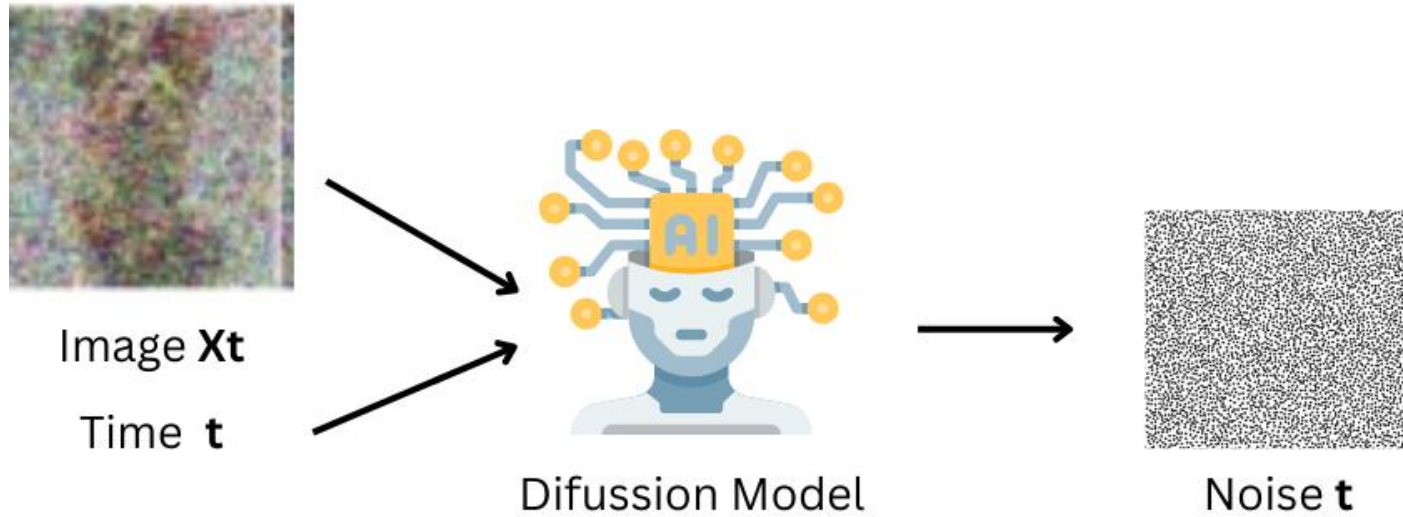


Difussion Model

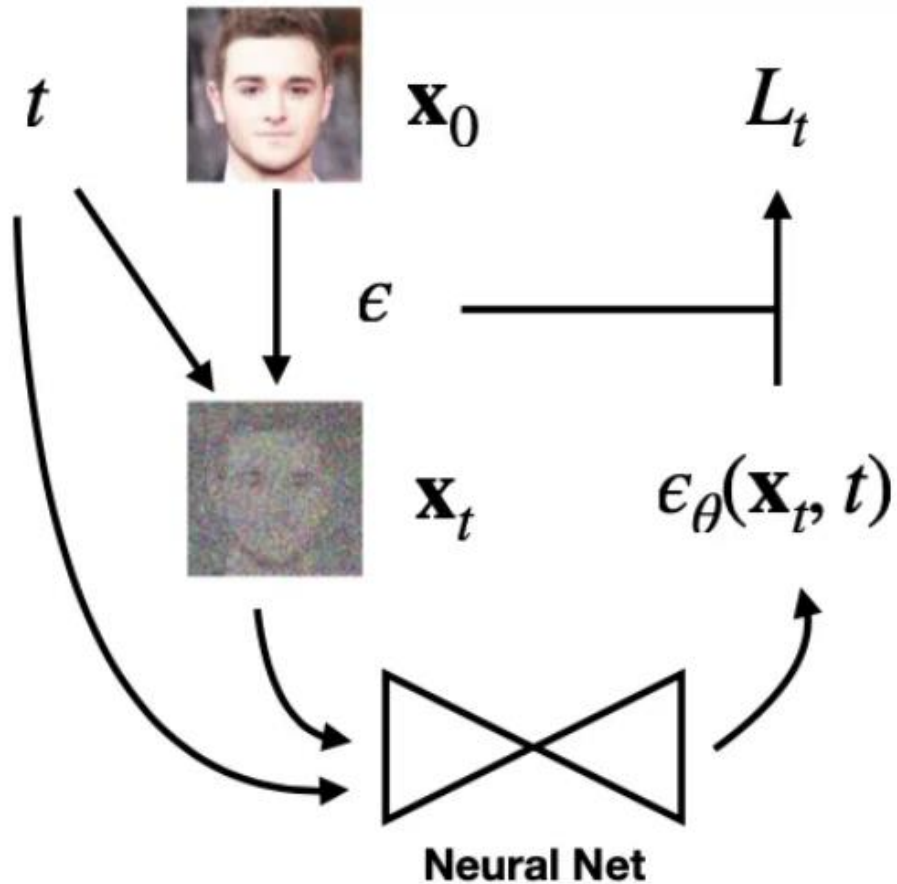


Noise t

Training Procedure



Training Procedure



Stable Diffusion Training Process. Image from <https://medium.com/@jjkoz762/diffusion-models-2d303735d219>

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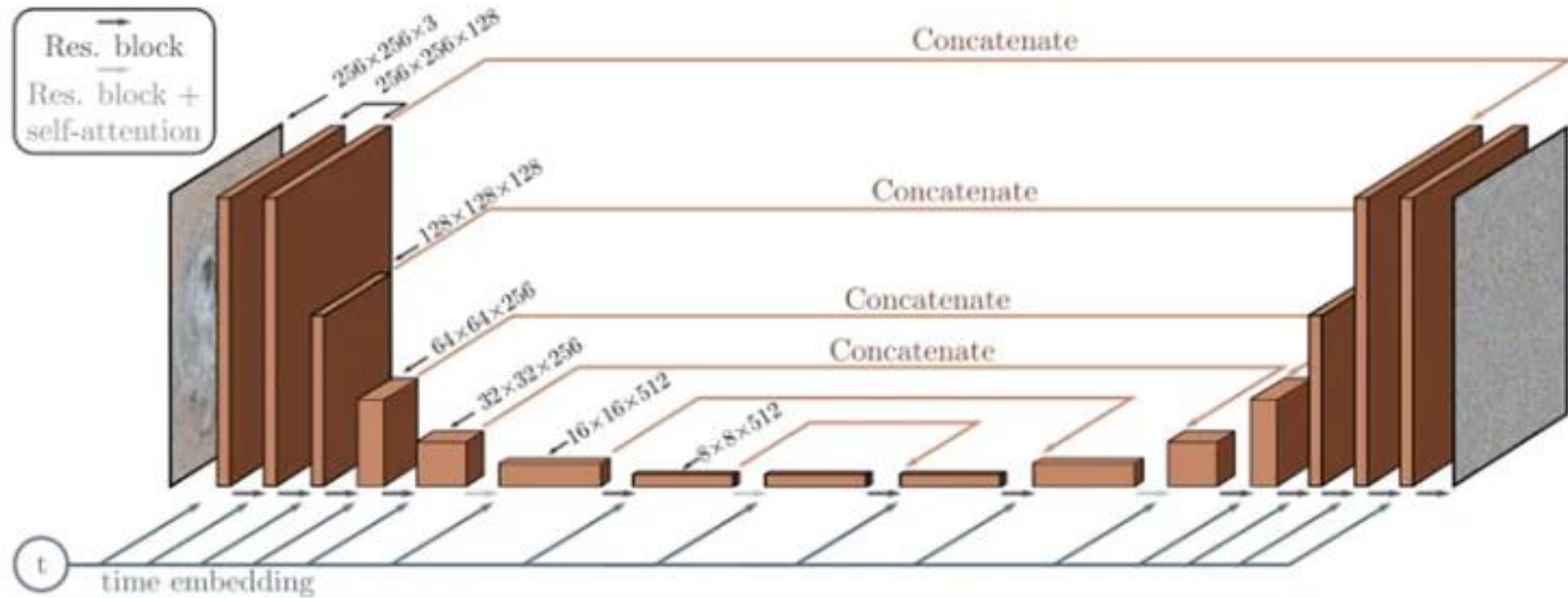
6718 8786



Or use QR code



Stable Diffusion – U-Net



U-Net^[11]. Image from <https://medium.com/@jjkoz762/diffusion-models-2d303735d219>

[11] Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18. Springer International Publishing, 2015.

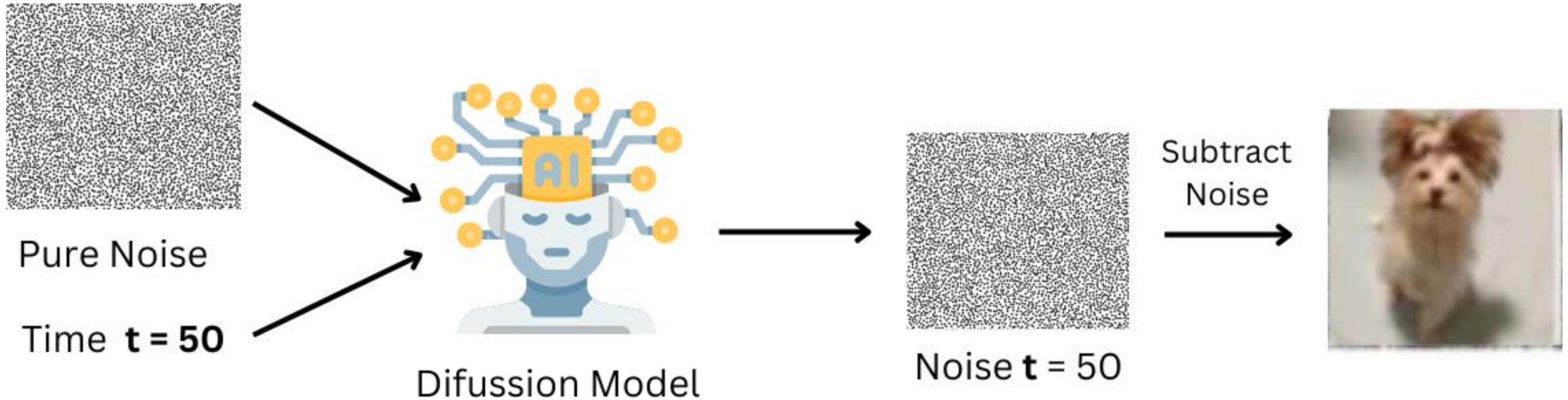
Stable Diffusion - Loss

$$L_{\text{simple}} = E_{t, x_0, \epsilon} [||\epsilon - \epsilon_{\theta}(x_t, t)||^2]$$

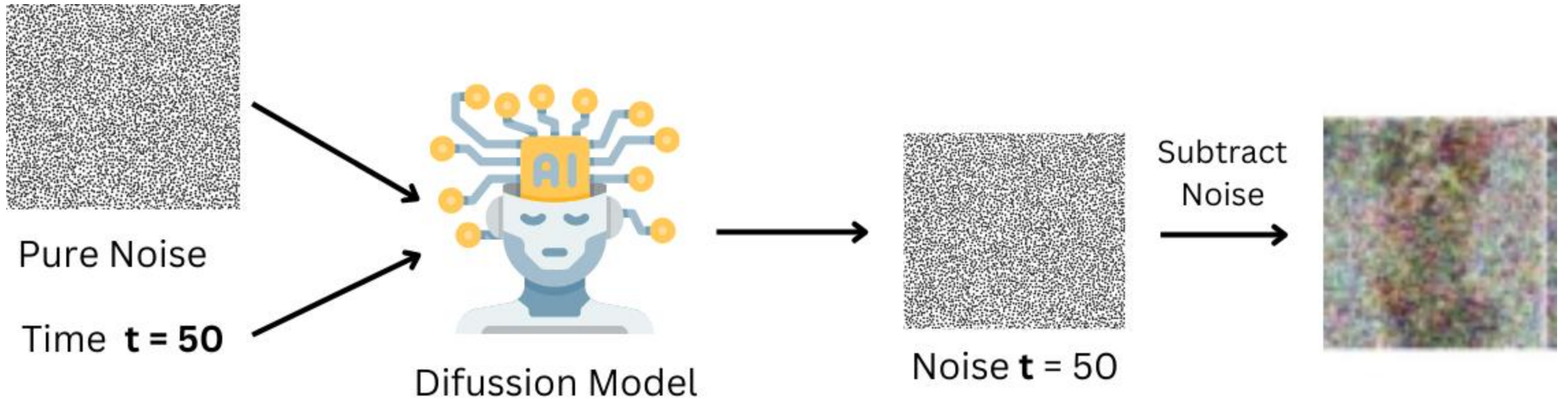
1. The loss from DDPM paper^[9]
2. Mean Squared Error between predicted noise and real noise

[9] Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." Advances in neural information processing systems 33 (2020): 6840-6851.

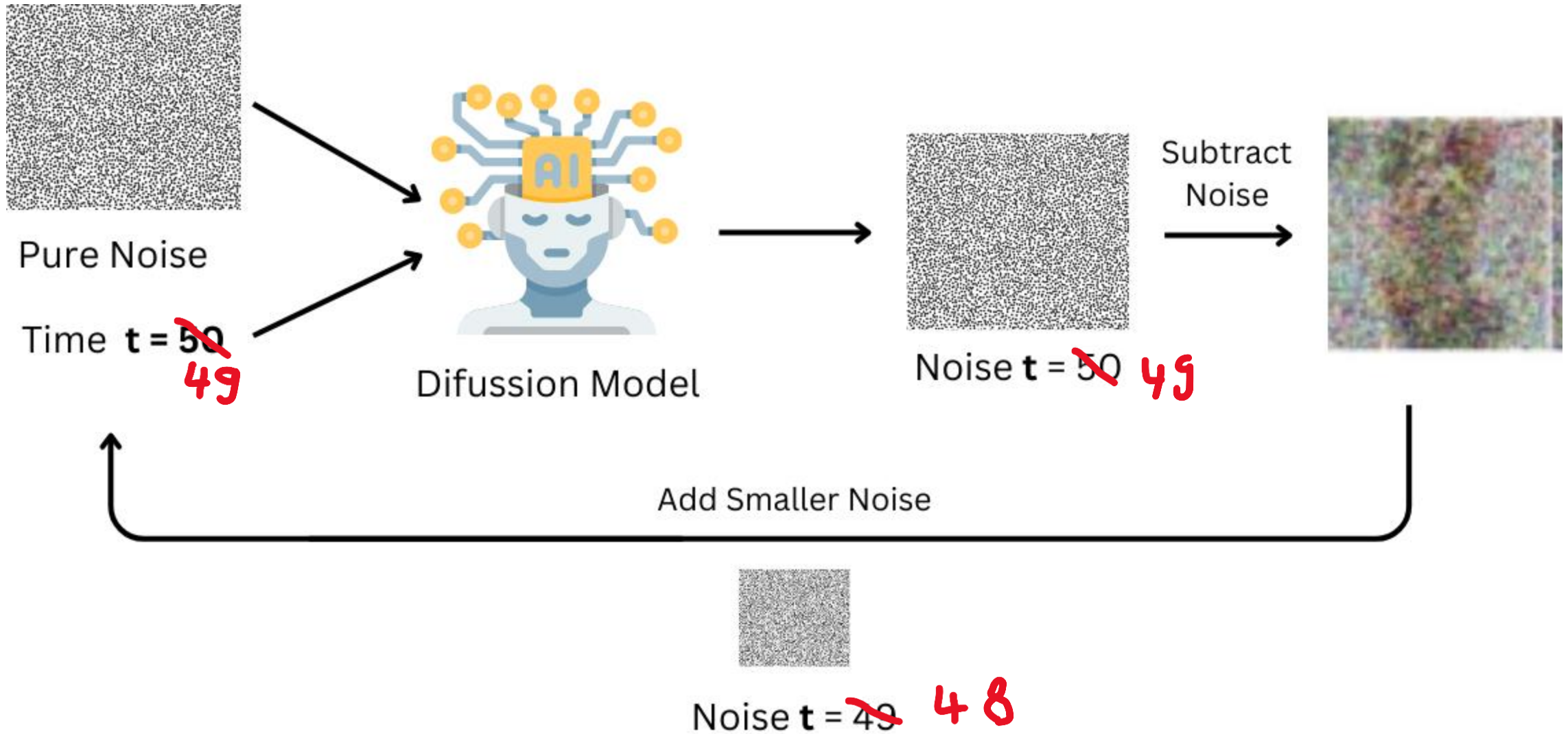
Generation Procedure



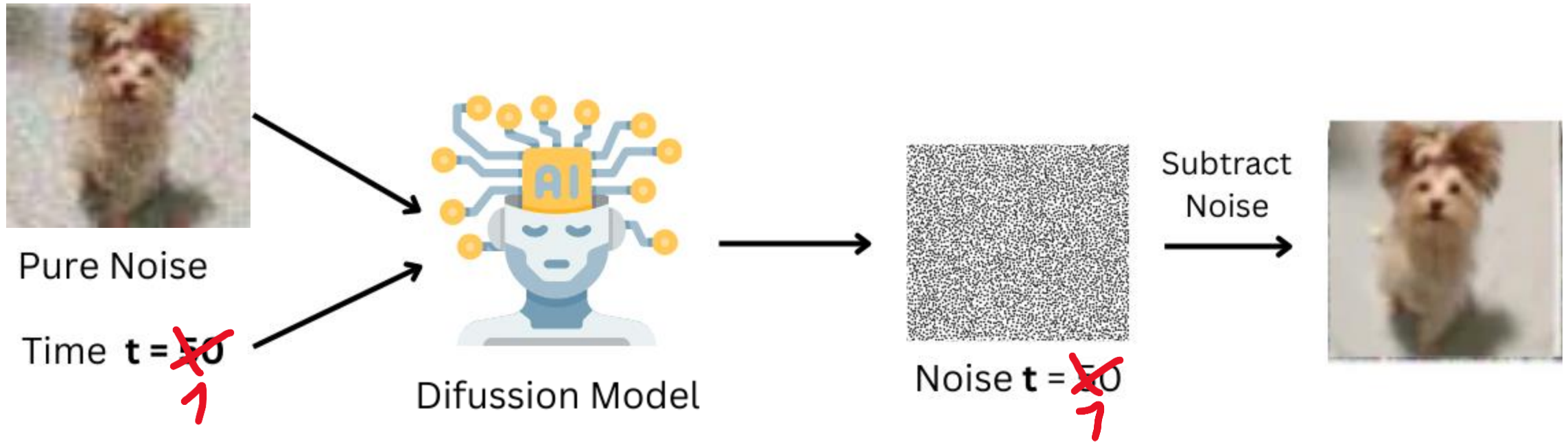
Generation Procedure



Generation Procedure



Generation Procedure



Computerphile Trick

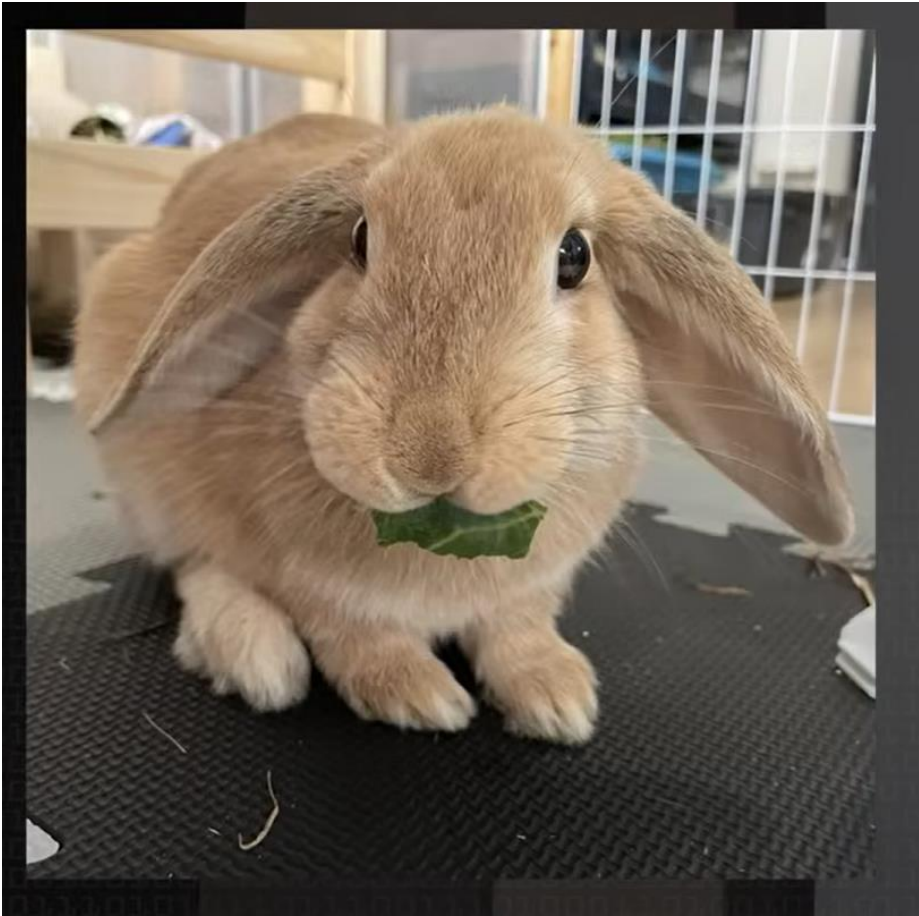


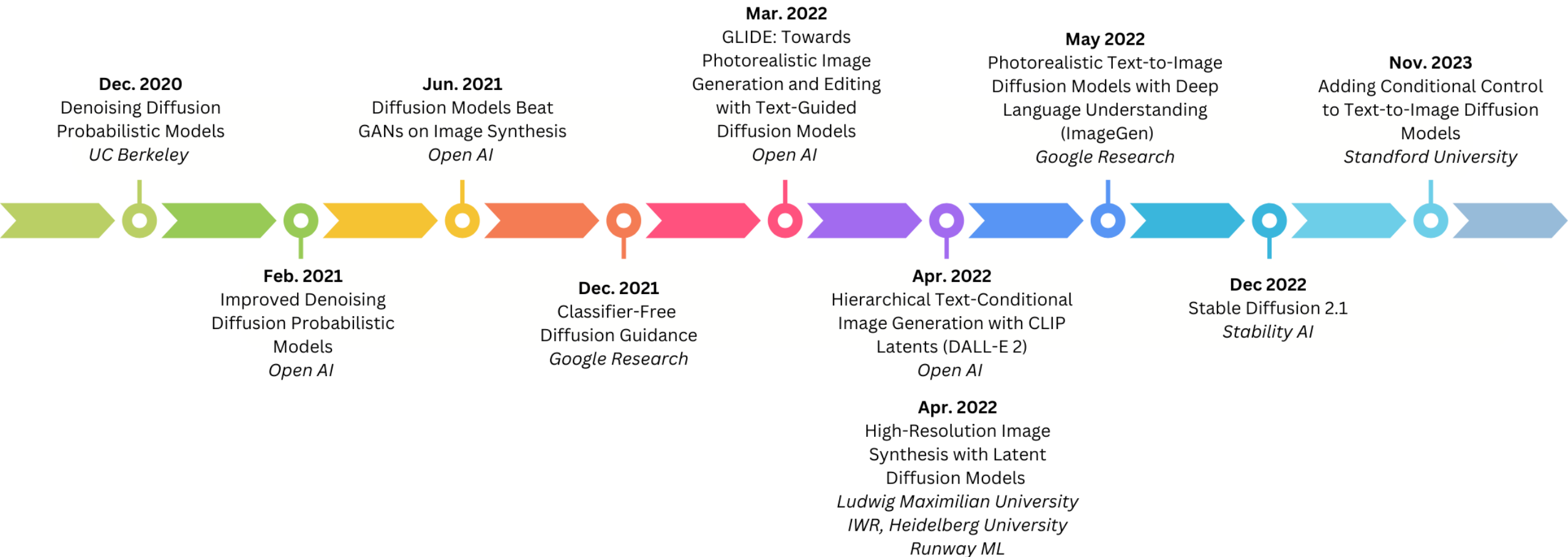
Image from “How I Understand Diffusion Models”- <https://www.youtube.com/watch?v=-lz30by8-sU>

Image Colorization

- Interesting Git Repo: <https://github.com/ErwannMillon/Color-diffusion>



Diffusion Timeline



Improved DDPM^[12]

- Learned Noise Schedules
 - Changed fixed variance -> learned (cosine) noise schedule
 - Reduced diffusions steps from 1000 to 50-100
- Classifier Conditioning

[12] Nichol, Alexander Quinn, and Prafulla Dhariwal. "Improved denoising diffusion probabilistic models." International conference on machine learning. PMLR, 2021.

Diffusion Models Beat GANs on Image Synthesis ^[13]

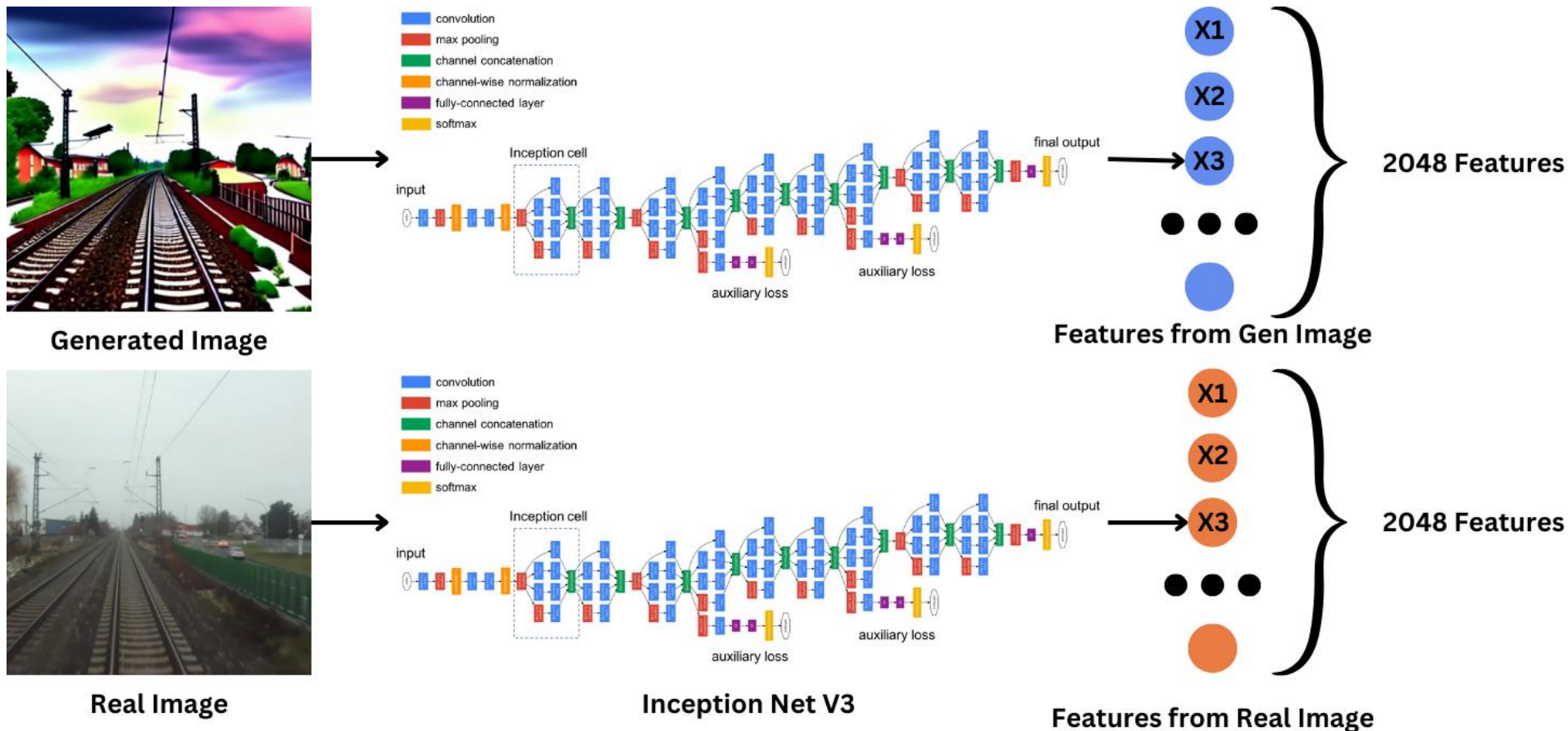
- Demonstrating Superiority of Diffusion over GANs
- Architecture Improvements. U-Net with self-attention
- Classifier Guidance
 - Required pre-trained classifier

[13] Dhariwal, Prafulla, and Alexander Nichol. "Diffusion models beat gans on image synthesis." Advances in neural information processing systems 34 (2021): 8780-8794.

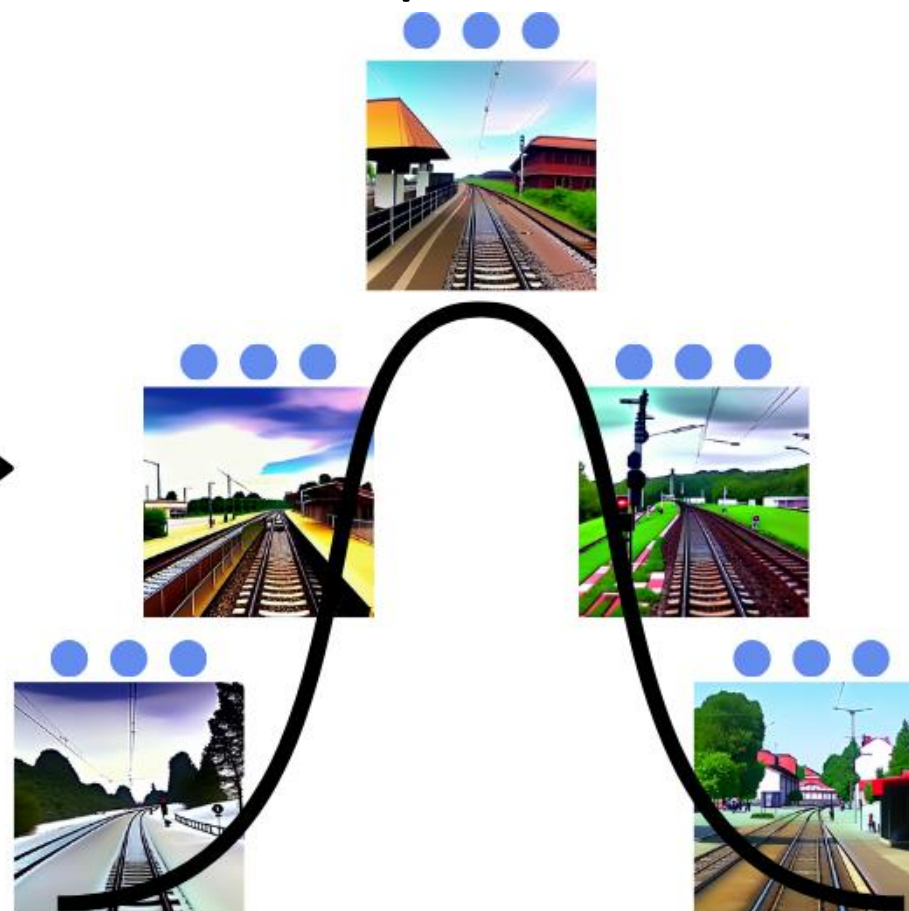


How do we evaluate
Image Generation?

Evaluation - Fréchet Inception Distance



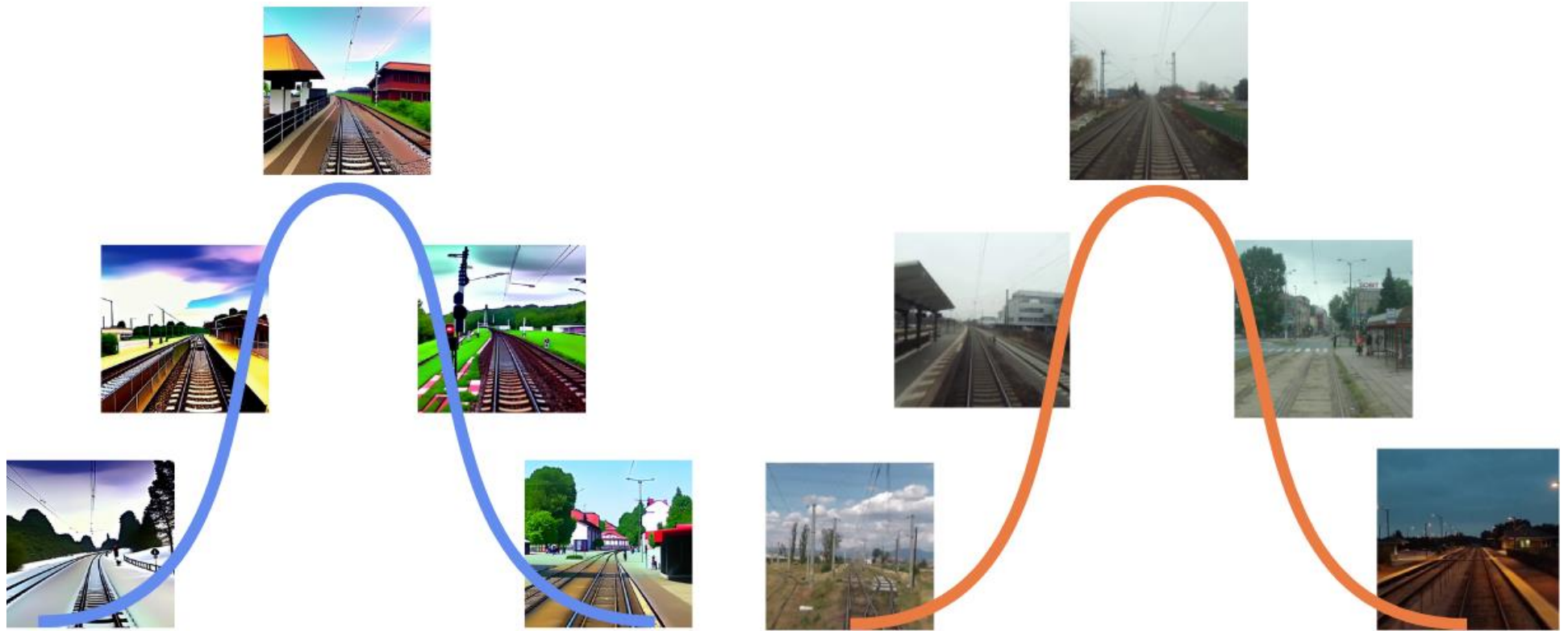
Evaluation - Fréchet Inception Distance



Features from Gen Image

Distribution of Generated Features

Evaluation - Fréchet Inception Distance

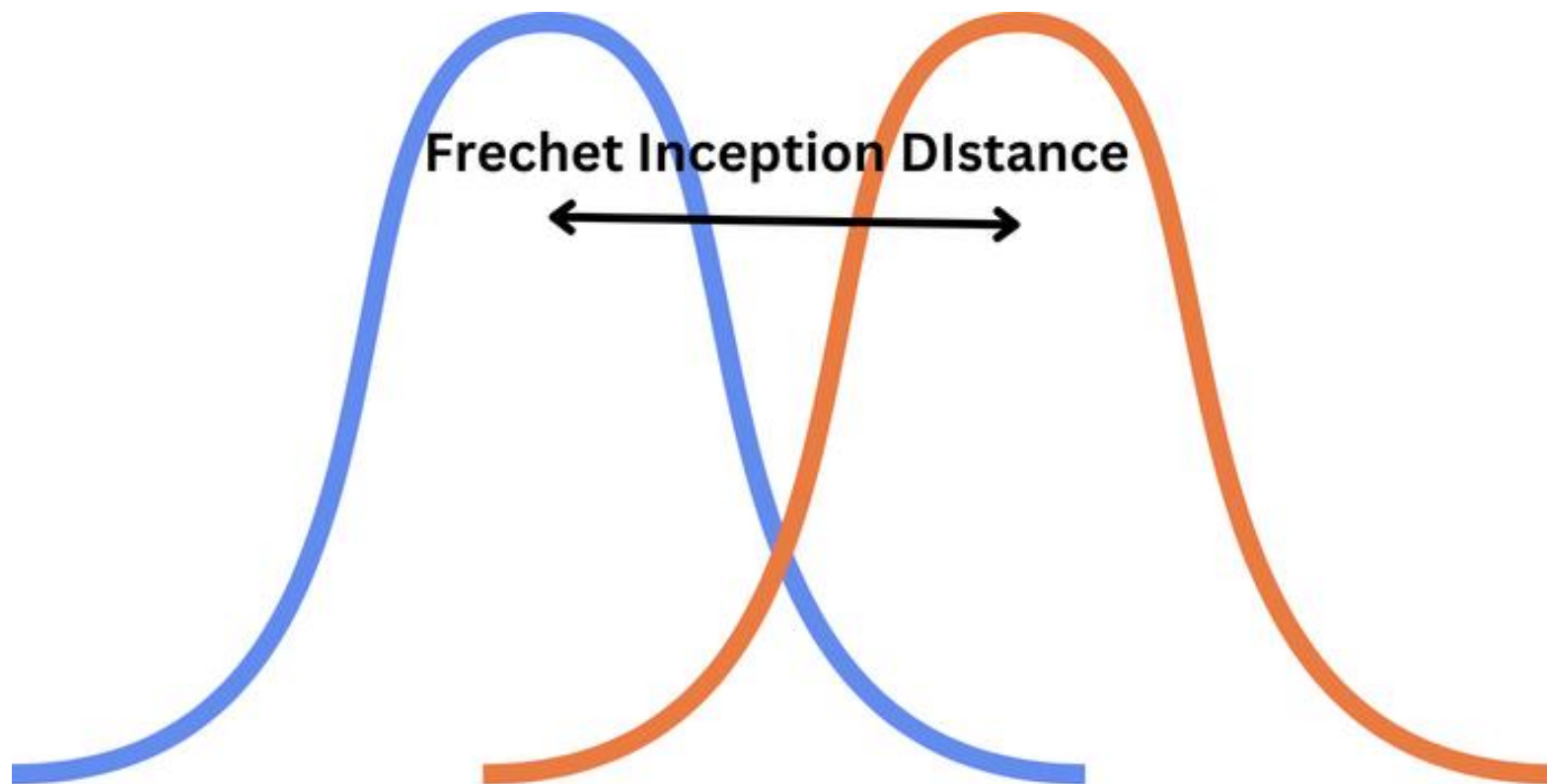


Distribution of Generated Features

Distribution of Real Features

Evaluation - Fréchet Inception Distance

$$FID = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$



Evaluation - Fréchet Inception Distance

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Wild Mentimeter
appeared!



Diffusion Models Beat GANs on Image Synthesis

Model	FID
LSUN Bedrooms 256×256	
DCTransformer [†] [42]	6.40
DDPM [25]	4.89
IDDPM [43]	4.24
StyleGAN [27]	2.35
ADM (dropout)	1.90
LSUN Horses 256×256	
StyleGAN2 [28]	3.84
ADM	2.95
ADM (dropout)	2.57
LSUN Cats 256×256	
DDPM [25]	17.1
StyleGAN2 [28]	7.25
ADM (dropout)	5.57
ImageNet 64×64	
BigGAN-deep* [5]	4.06
IDDPM [43]	2.92
ADM	2.61
ADM (dropout)	2.07

Model	FID
ImageNet 128×128	
BigGAN-deep [5]	6.02
LOGAN [†] [68]	3.36
ADM	5.91
ADM-G (25 steps)	5.98
ADM-G	2.97
ImageNet 256×256	
DCTransformer [†] [42]	36.51
VQ-VAE-2 ^{†‡} [51]	31.11
IDDPM [‡] [43]	12.26
SR3 ^{†‡} [53]	11.30
BigGAN-deep [5]	6.95
ADM	10.94
ADM-G (25 steps)	5.44
ADM-G	4.59
ImageNet 512×512	
BigGAN-deep [5]	8.43
ADM	23.24
ADM-G (25 steps)	8.41
ADM-G	7.72

Diffusion Models Beat GANs on Image Synthesis

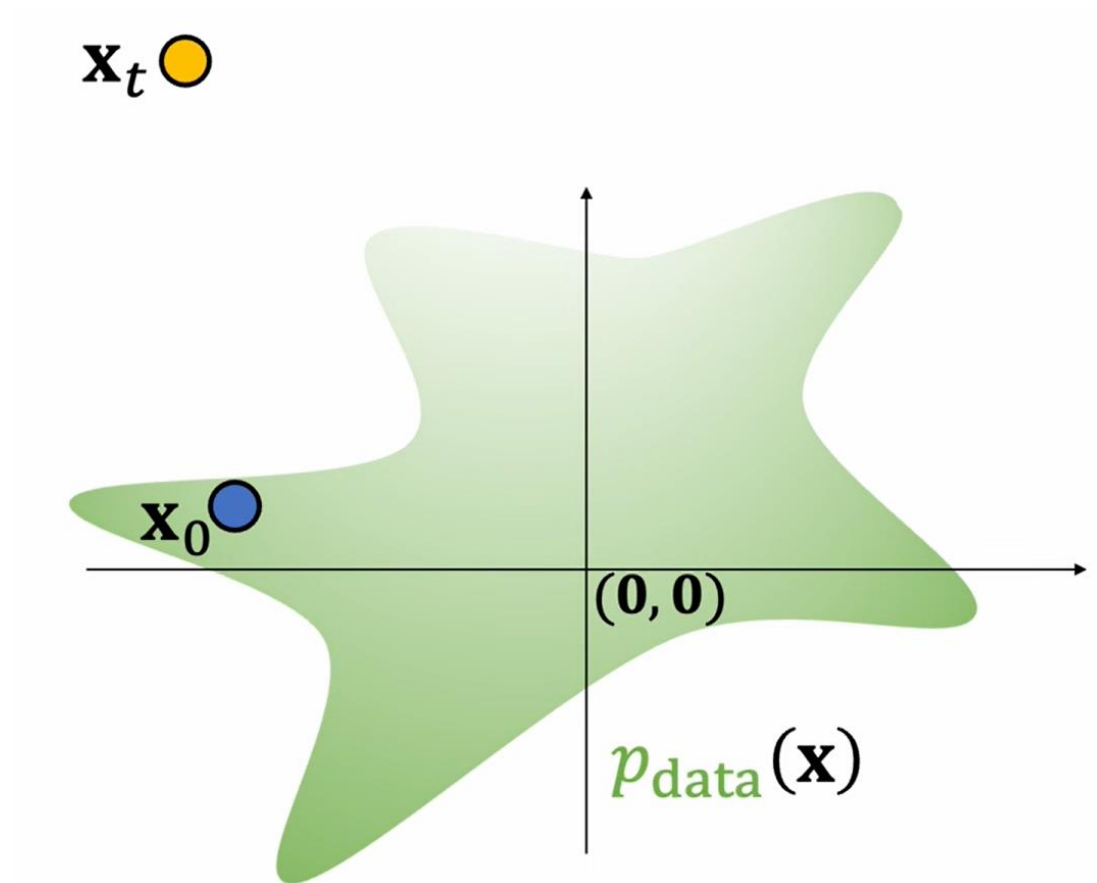


Image from “How I Understand Diffusion Models” - <https://www.youtube.com/watch?v=i2qSxMVeVLI>

Diffusion Models Beat GANs on Image Synthesis

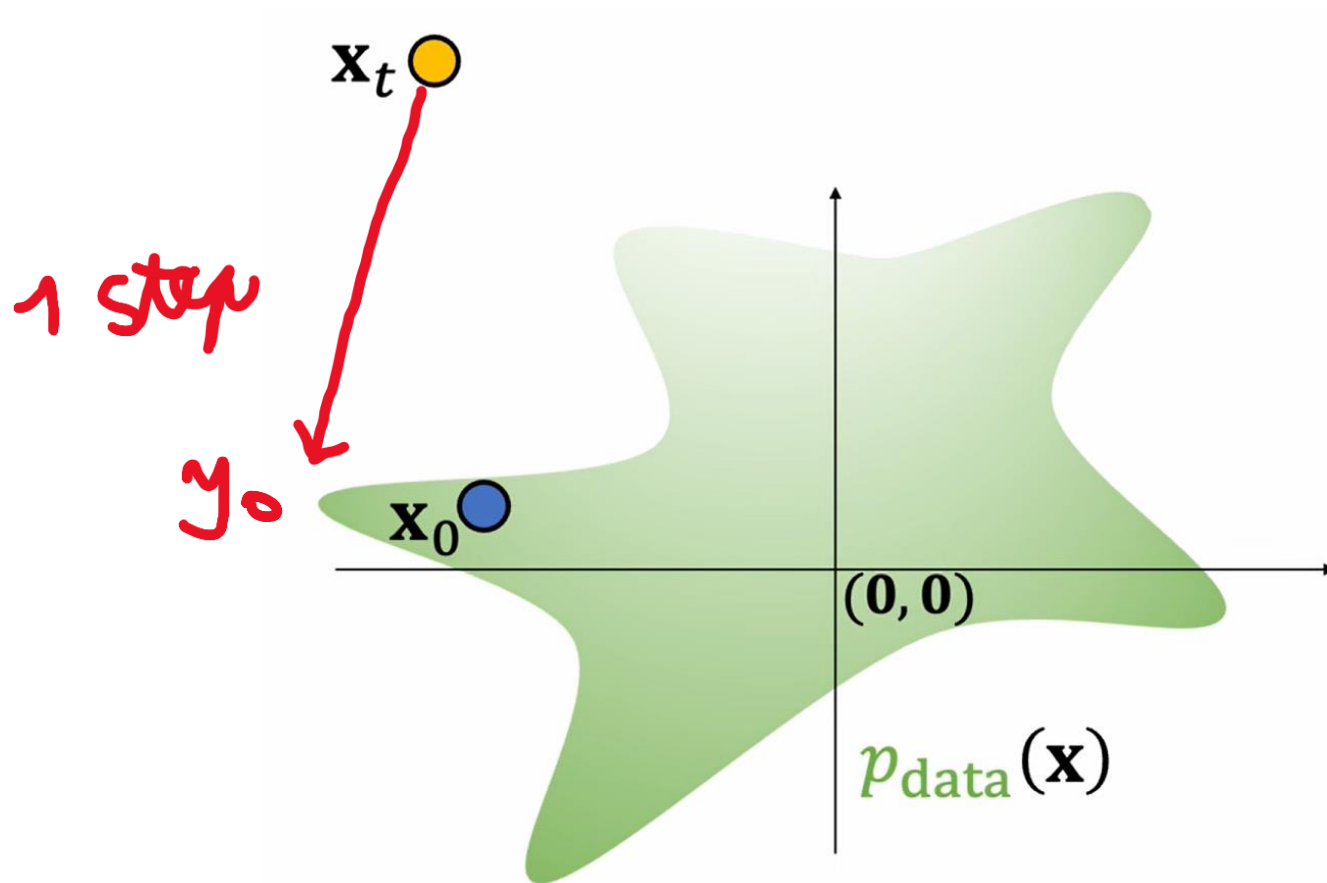


Image from "How I Understand Diffusion Models" - <https://www.youtube.com/watch?v=i2qSxMVeVLI>

Diffusion Models Beat GANs on Image Synthesis

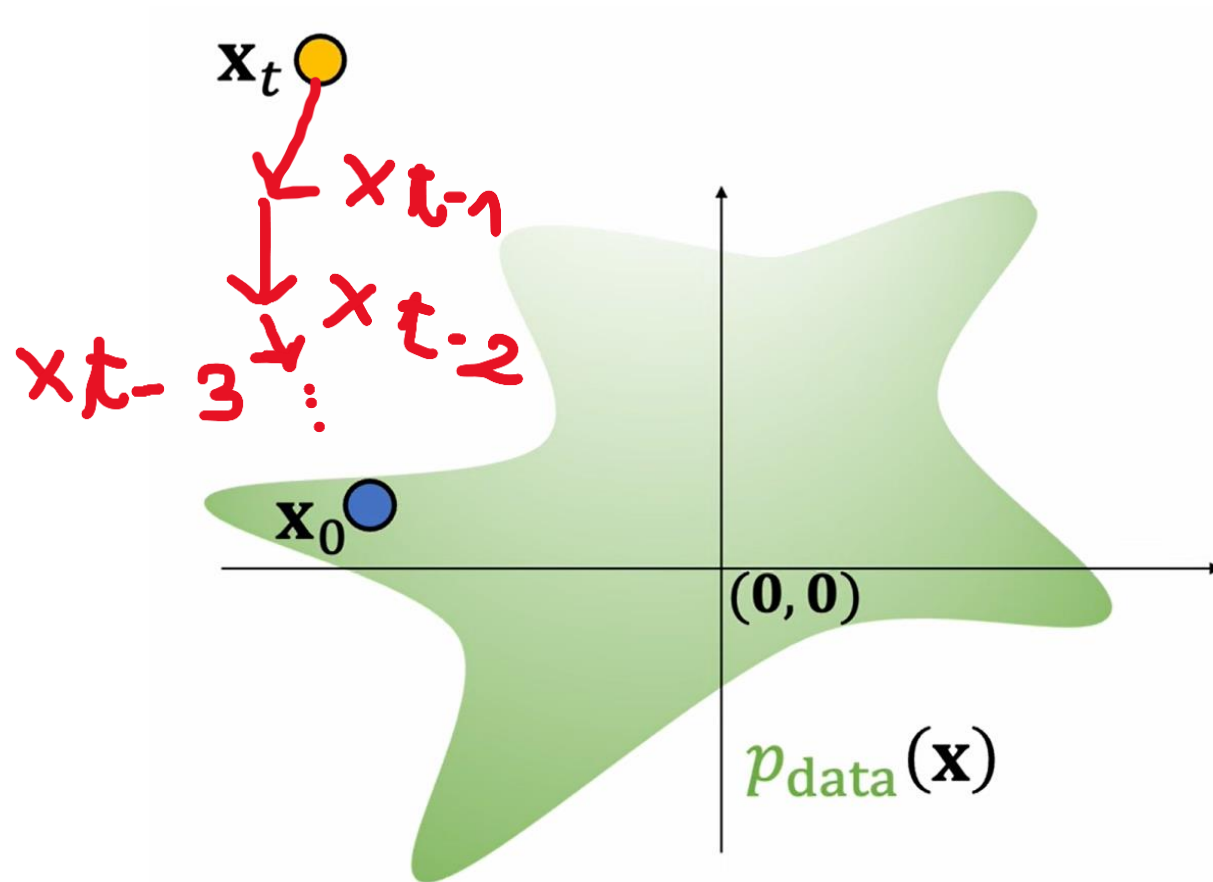


Image from “How I Understand Diffusion Models” - <https://www.youtube.com/watch?v=i2qSxMVeVLI>

Classifier Guidance

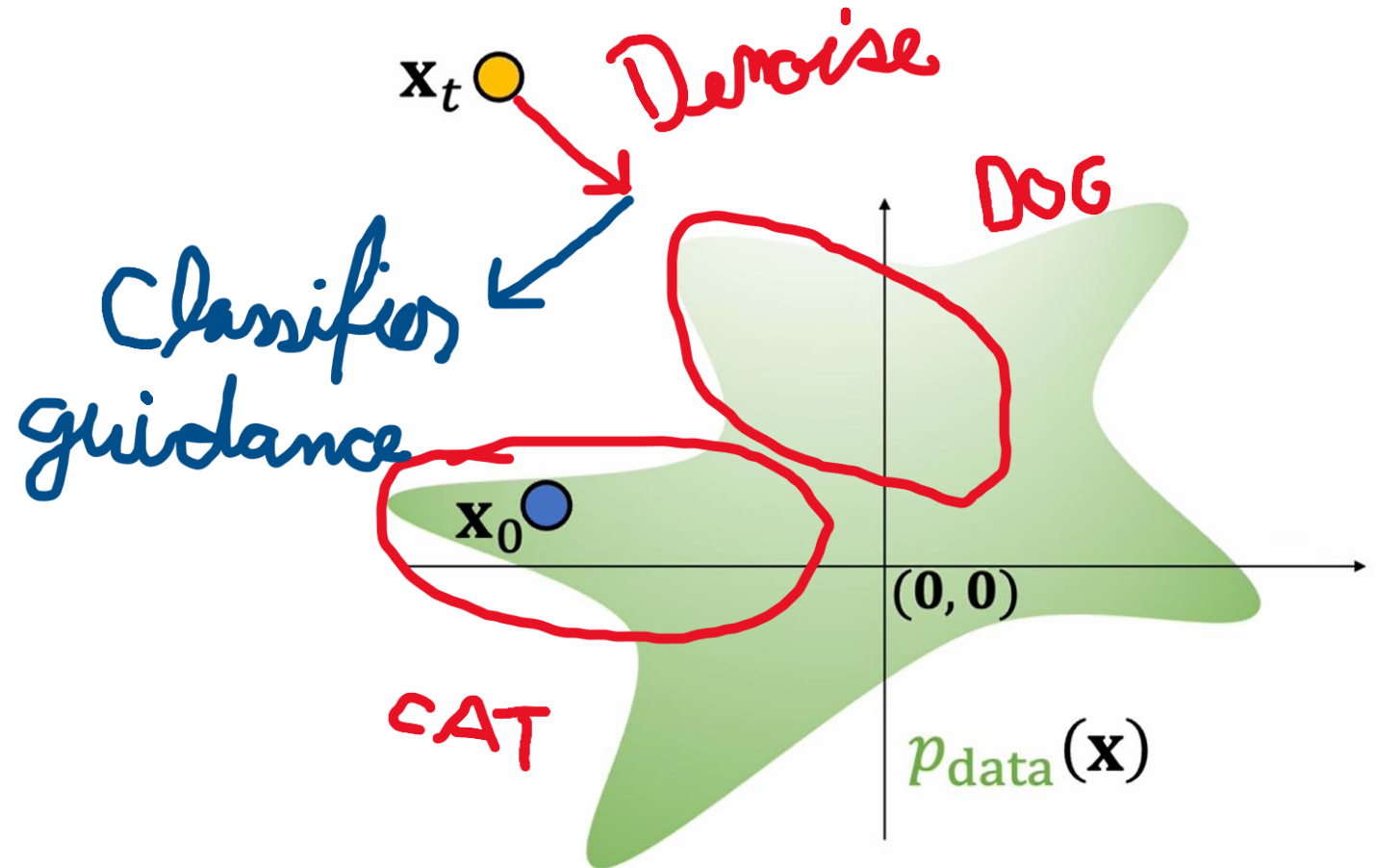


Image from "How I Understand Diffusion Models" - <https://www.youtube.com/watch?v=i2qSxMVeVLI>

Classifier-Free Diffusion Guidance [14]

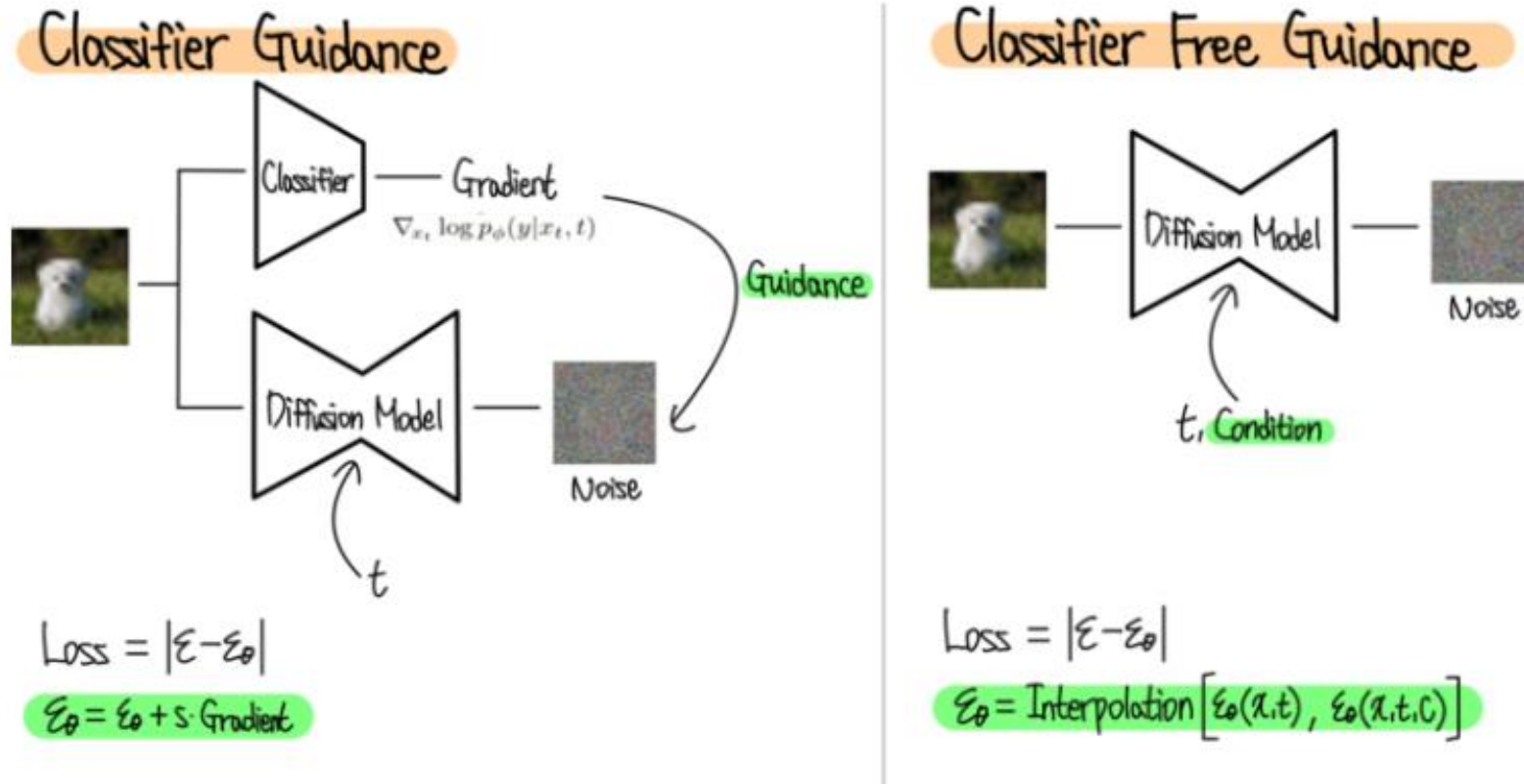


Image from <https://ffighting.net/deep-learning-paper-review/diffusion-model/classifier-free-guidance/>

GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models ^[15]

- Text-Guided Image Generation
- CLIP ^[16] guidance
- Classifier guidance yielded better results

[15] Nichol, Alex, et al. "Glide: Towards photorealistic image generation and editing with text-guided diffusion models." arXiv preprint arXiv:2112.10741 (2021).

[16] Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.

CLIP

(1) Contrastive pre-training

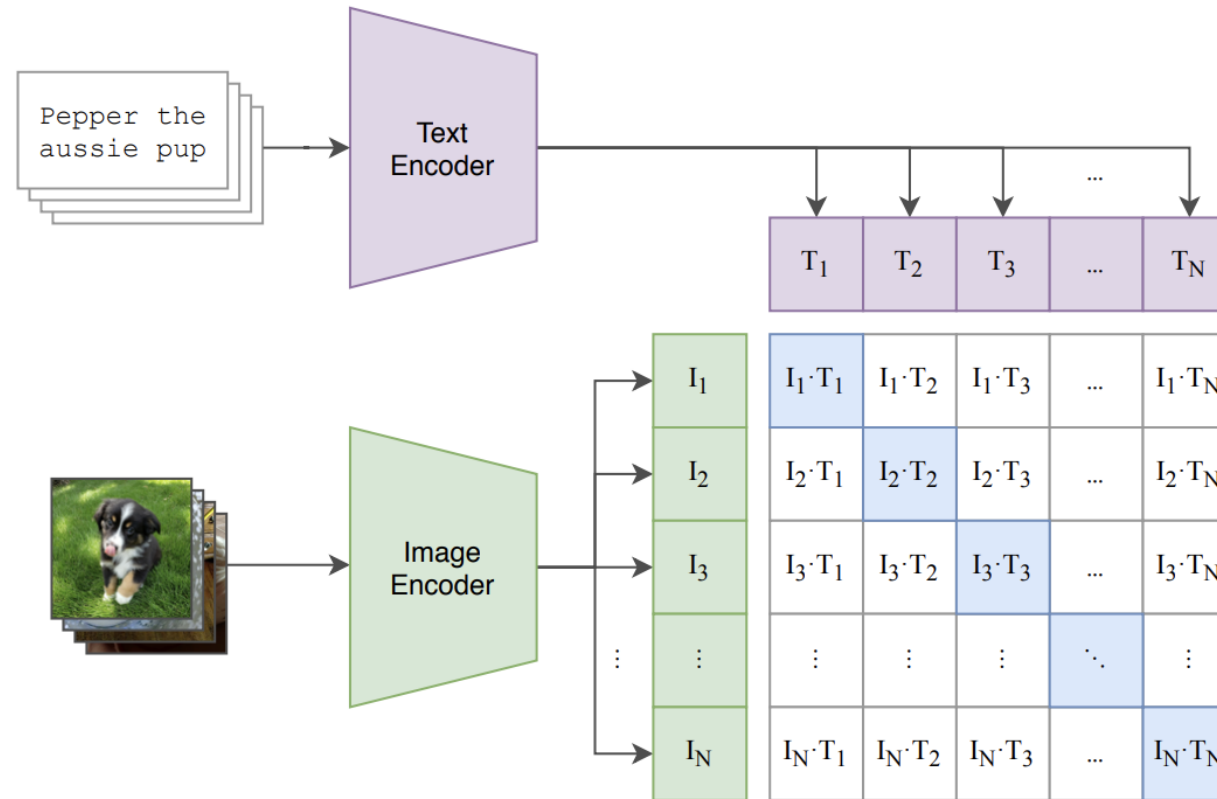


Image from Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.



"a hedgehog using a calculator"



"a corgi wearing a red bowtie and a purple party hat"



"robots meditating in a vipassana retreat"



"a fall landscape with a small cottage next to a lake"



"a surrealist dream-like oil painting by salvador dali of a cat playing checkers"



"a professional photo of a sunset behind the grand canyon"



"a high-quality oil painting of a psychedelic hamster dragon"



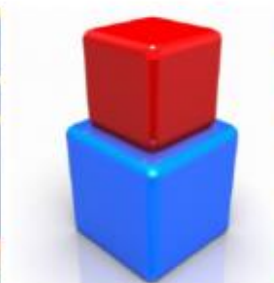
"an illustration of albert einstein wearing a superhero costume"



"a boat in the canals of venice"



"a painting of a fox in the style of starry night"



"a red cube on top of a blue cube"



"a stained glass window of a panda eating bamboo"



"a crayon drawing of a space elevator"



"a futuristic city in synthwave style"



"a pixel art corgi pizza"



"a fog rolling into new york"

Hierarchical Text-Conditional Image Generation with CLIP Latents (UnCLIP \ DALL-E 2) ^[17]

- Text embeddings of CLIP (as opposed to transfer text embeddings)
- Feed image encodings to the decoder

UnCLIP \ DALL-E 2

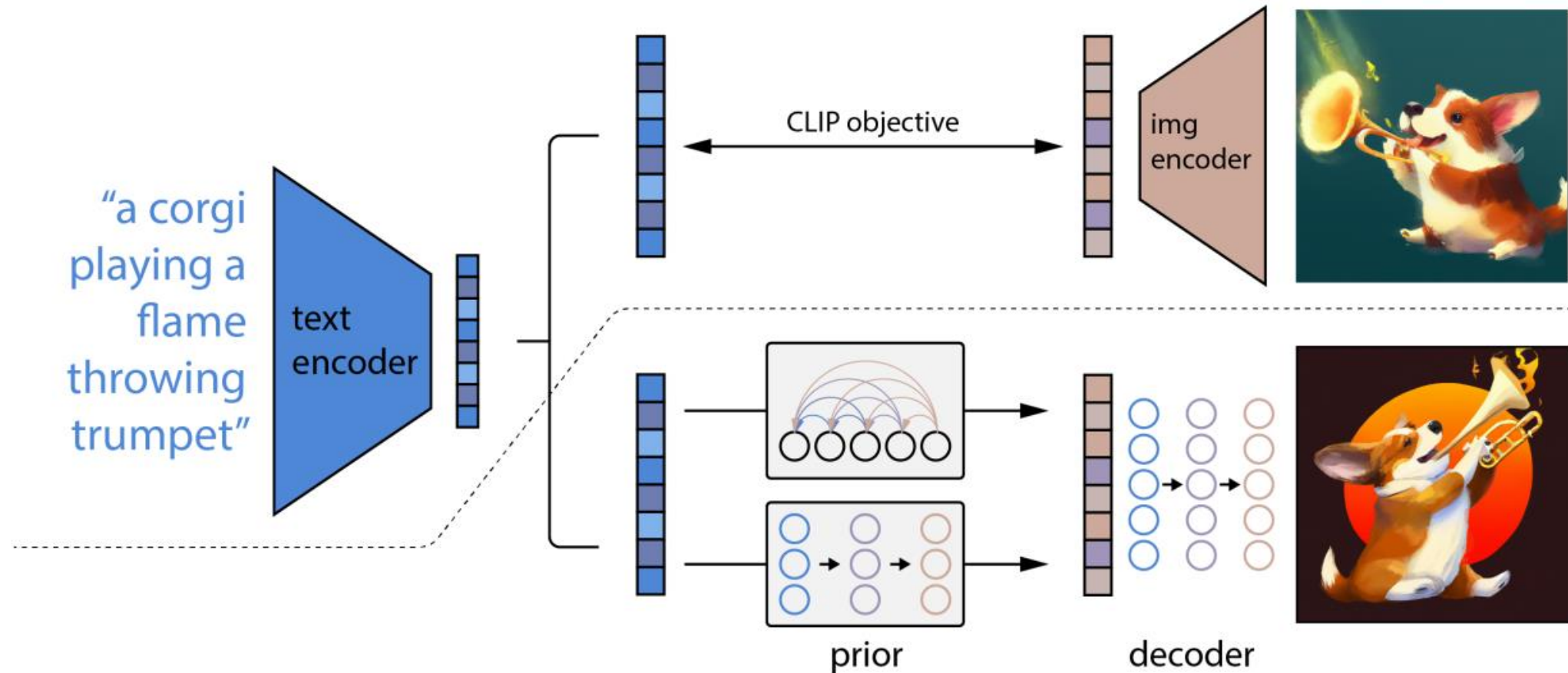
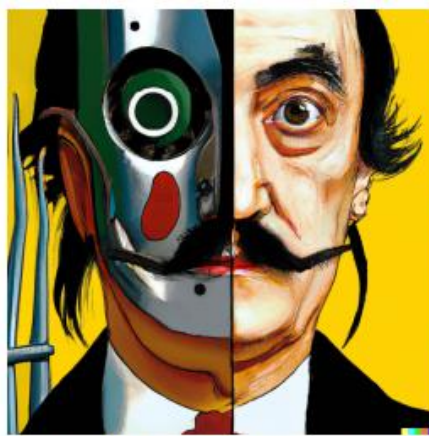


Image from Ramesh, Aditya, et al. "Hierarchical text-conditional image generation with clip latents." arXiv preprint arXiv:2204.06125 1.2 (2022): 3..



a vibrant portrait painting of Salvador Dalí with a robotic half face



a shiba inu wearing a beret and black turtleneck



a close up of a handpalm with leaves growing from it



an espresso machine that makes coffee from human souls, artstation



panda mad scientist mixing sparkling chemicals, artstation



a corgi's head depicted as an explosion of a nebula



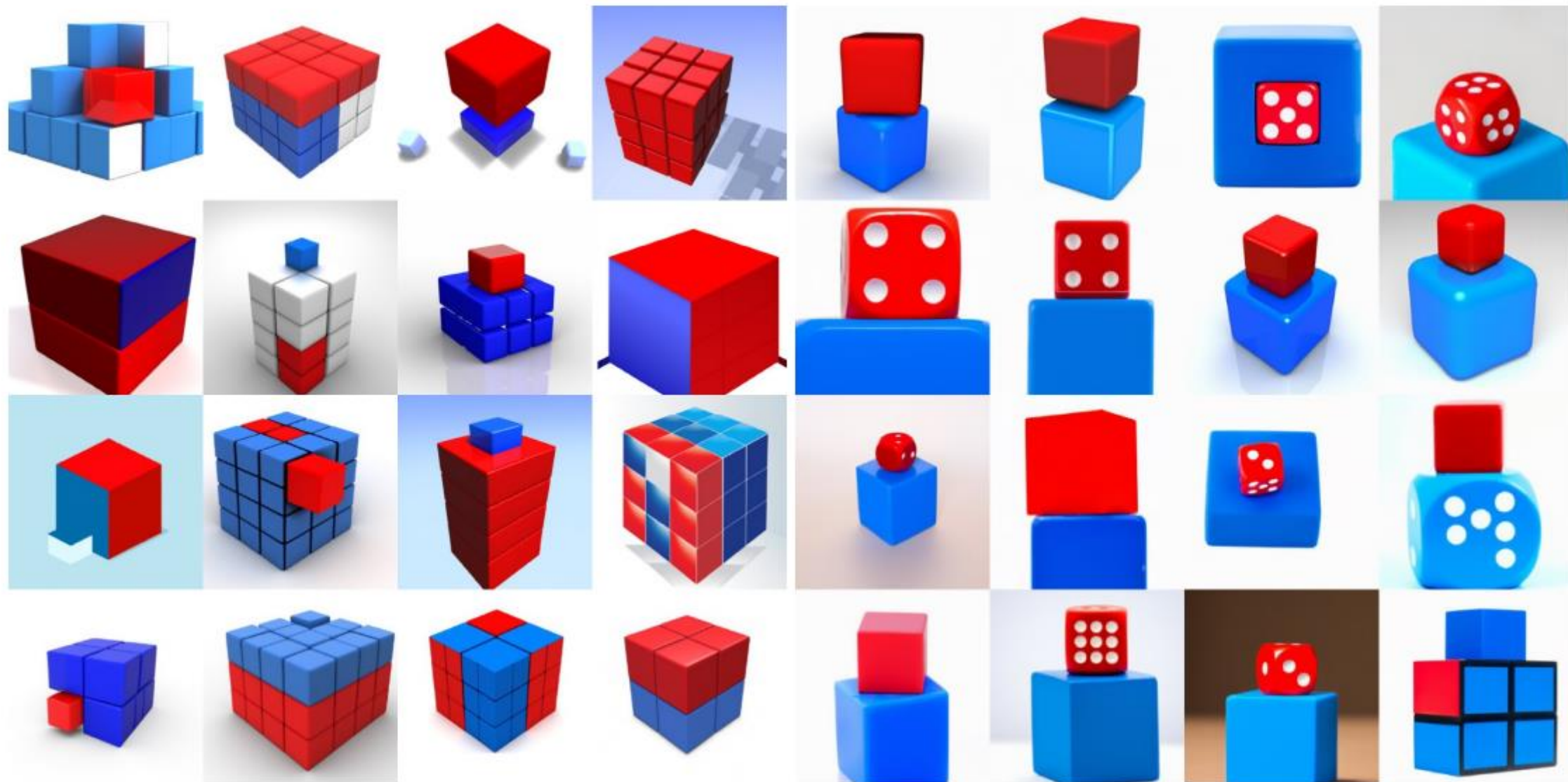
a dolphin in an astronaut suit on saturn, artstation



a propaganda poster depicting a cat dressed as french emperor napoleon holding a piece of cheese



a teddy bear on a skateboard in times square



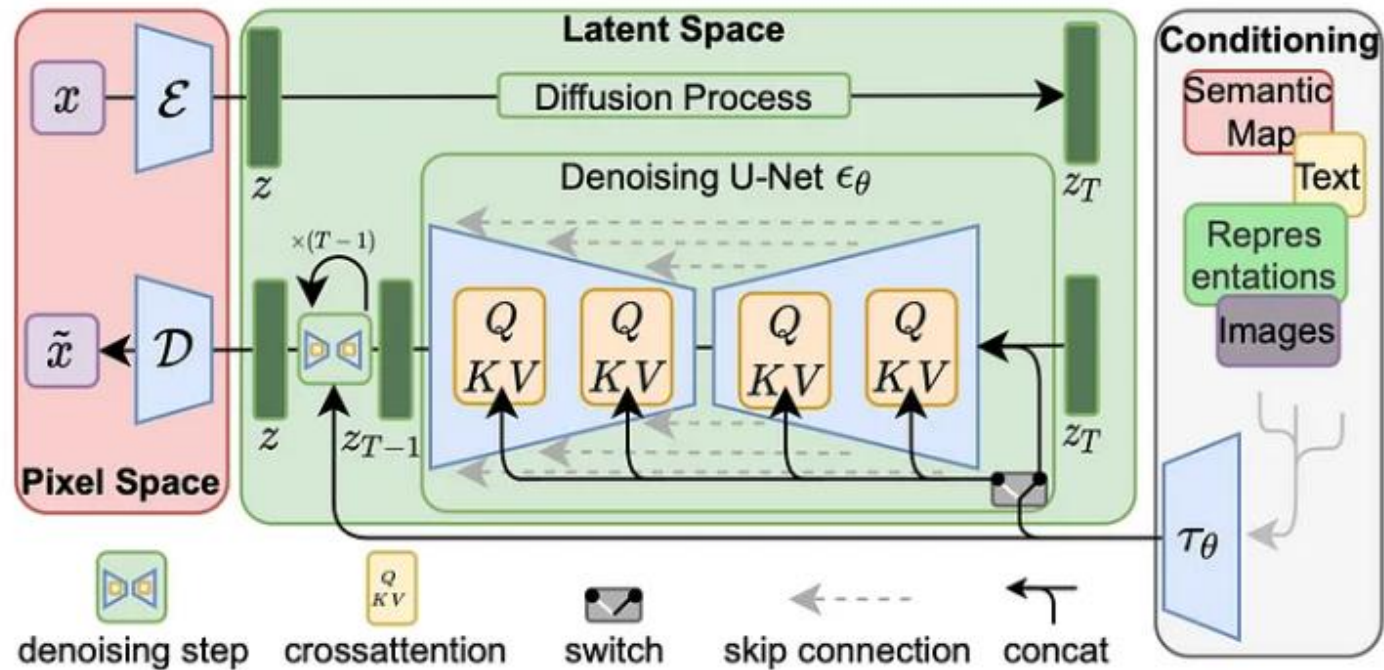
(a) unCLIP

(b) GLIDE

Image from Ramesh, Aditya, et al. "Hierarchical text-conditional image generation with clip latents." arXiv preprint arXiv:2204.06125 1.2 (2022): 3..

High-Resolution Image Synthesis with Latent Diffusion Models ^[18]

- Diffusion in Latent Space Instead of Pixel Space
- LDM allow high-resolution image synthesis (1024x1024)



[18] Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022.



Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding^[19]

- Google ImageGen: <https://imagen.research.google/>
- Enhanced Photorealism
- Advanced Language Understanding (PaLM)
- Multiscale Diffusion

[19] Saharia, Chitwan, et al. "Photorealistic text-to-image diffusion models with deep language understanding." Advances in neural information processing systems 35 (2022): 36479-36494.

Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding^[19]

Model	FID-30K	Zero-shot FID-30K
AttnGAN [76]	35.49	
DM-GAN [83]	32.64	
DF-GAN [69]	21.42	
DM-GAN + CL [78]	20.79	
XMC-GAN [81]	9.33	
LAFITE [82]	8.12	
Make-A-Scene [22]	7.55	
DALL-E [53]		17.89
LAFITE [82]		26.94
GLIDE [41]		12.24
DALL-E 2 [54]		10.39
Imagen (Our Work)		7.27



Sprouts in the shape of text 'Imagen' coming out of a fairytale book.



A photo of a Shiba Inu dog with a backpack riding a bike. It is wearing sunglasses and a beach hat.



A high contrast portrait of a very happy fuzzy panda dressed as a chef in a high end kitchen making dough. There is a painting of flowers on the wall behind him.



Teddy bears swimming at the Olympics 400m Butterfly event.



A cute corgi lives in a house made out of sushi.



A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.



A brain riding a rocketship heading towards the moon.



A dragon fruit wearing karate belt in the snow.



A strawberry mug filled with white sesame seeds. The mug is floating in a dark chocolate sea.

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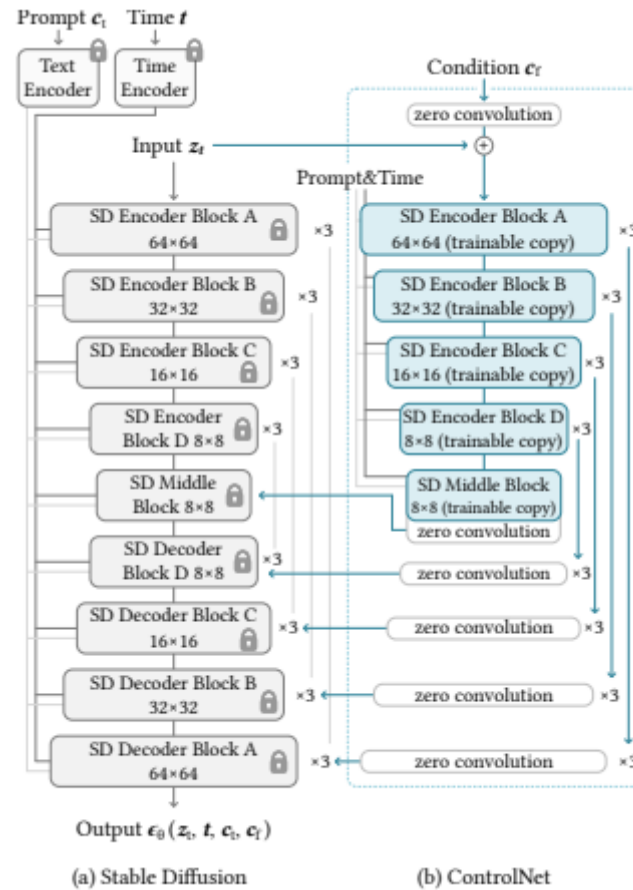
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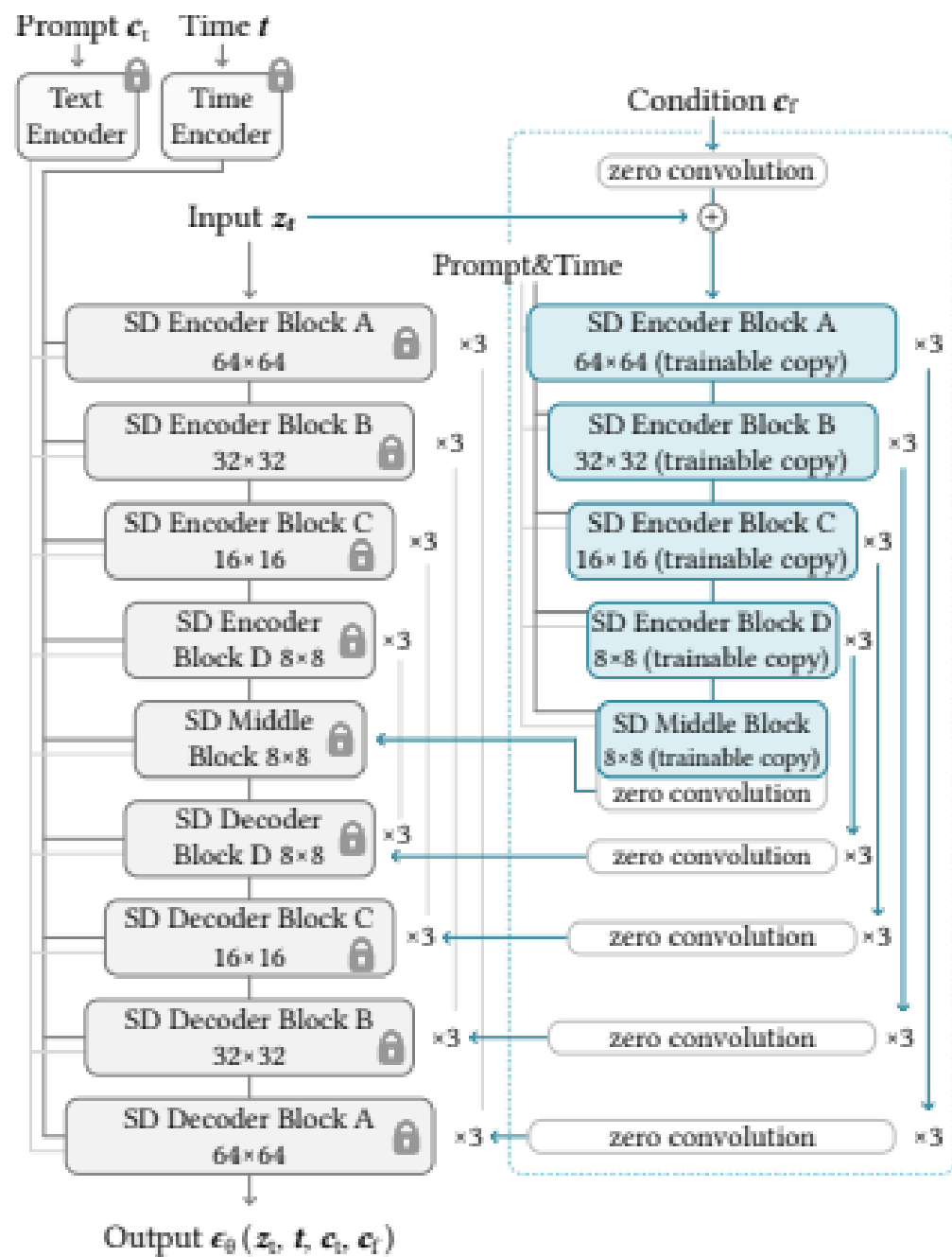
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Adding Conditional Control to Text-to-Image Diffusion Models [20]



[20] Zhang, Lvmin, Anyi Rao, and Maneesh Agrawala. "Adding conditional control to text-to-image diffusion models." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.



(a) Stable Diffusion

(b) ControlNet



Input Canny edge



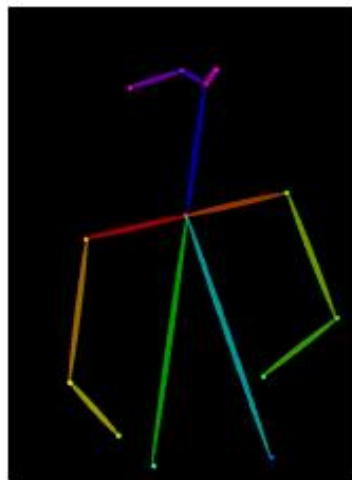
Default



“masterpiece of fairy tale, giant deer, golden antlers”



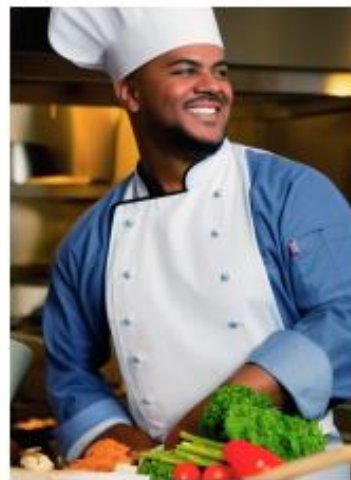
“..., quaint city Galic”



Input human pose



Default



“chef in kitchen”



“Lincoln statue”

Other recent advancements

- DALL-E 3 (late 2023) – Multi- Modal
- Diffusion Transformer Models
 - SORA – Video Generation: <https://openai.com/index/sora/>
 - Stable Diffusion 3 – Stability AI
- ImageGen 3
- Control Net Variants
- DeepCache: Accelerating Diffusion Models for Free ^[21]

[21] Ma, Xinyin, Gongfan Fang, and Xinchao Wang. "Deepcache: Accelerating diffusion models for free." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.

Final Question

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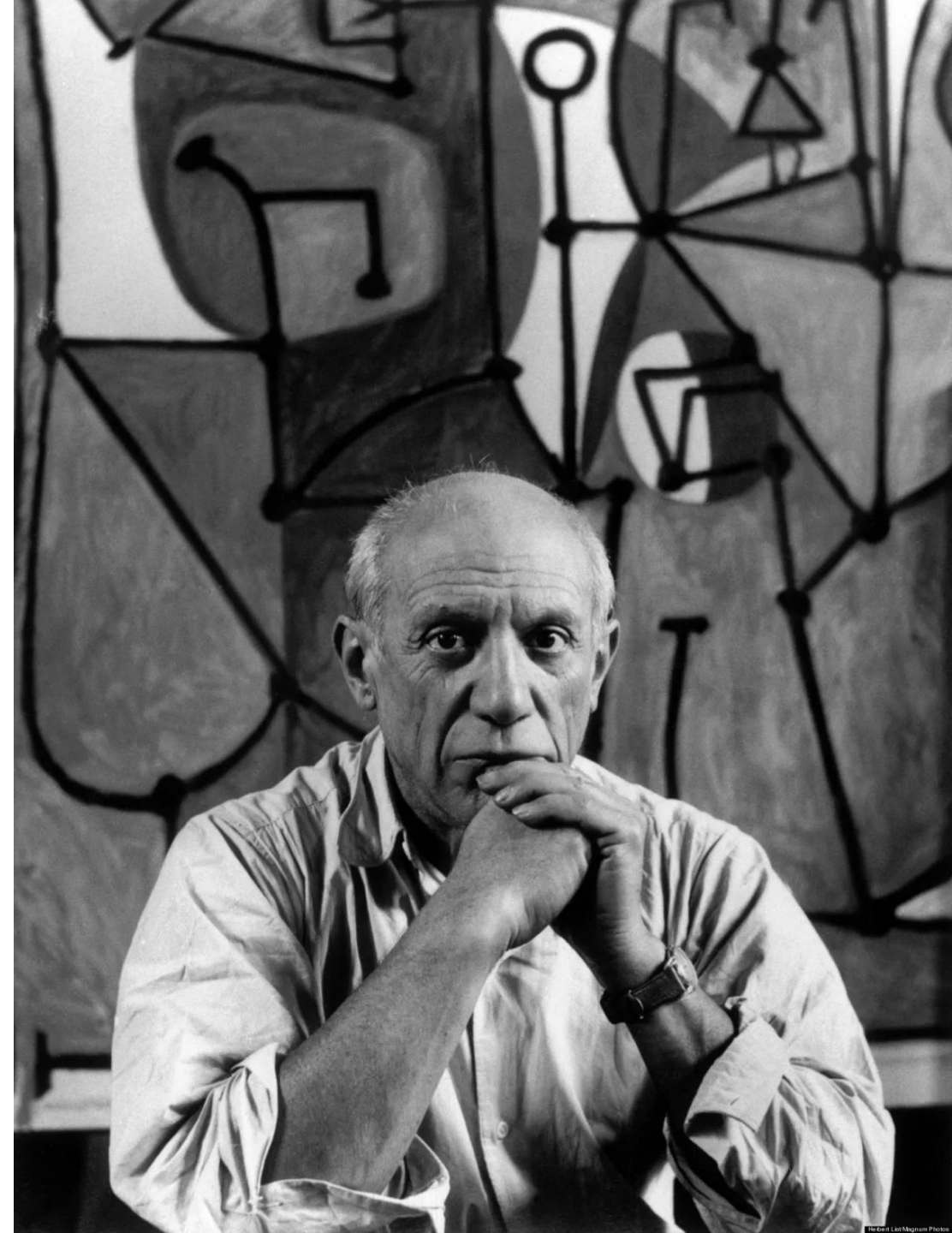


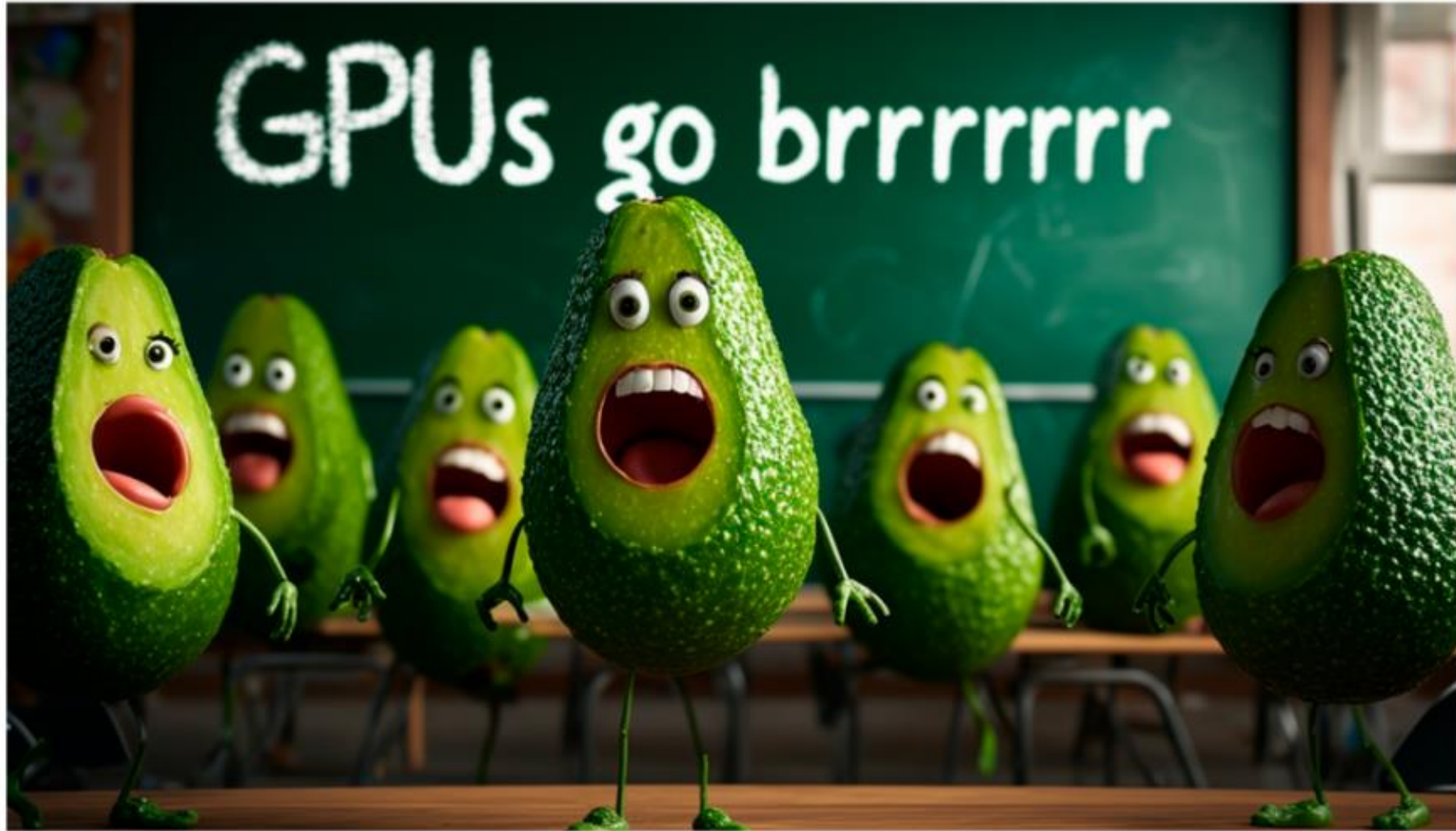
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“Good artists copy,
great artists steal.”

- *Pablo Picasso*

Question: Are Stable
Diffusion models good or
great artists? Do they
copy or do they steal?





Prompt: A surreal and humorous scene in a classroom with the words 'GPUs go brrrrrr' written in white chalk on a blackboard. In front of the blackboard, a group of students are celebrating. These students are uniquely depicted as avocados, complete with little arms and legs, and faces showing expressions of joy and excitement. The scene captures a playful and imaginative atmosphere, blending the concept of a traditional classroom with the whimsical portrayal of avocado students.

Image from Stability AI – Stable Diffusion 3 <https://stability.ai/news/stable-diffusion-3-research-paper>